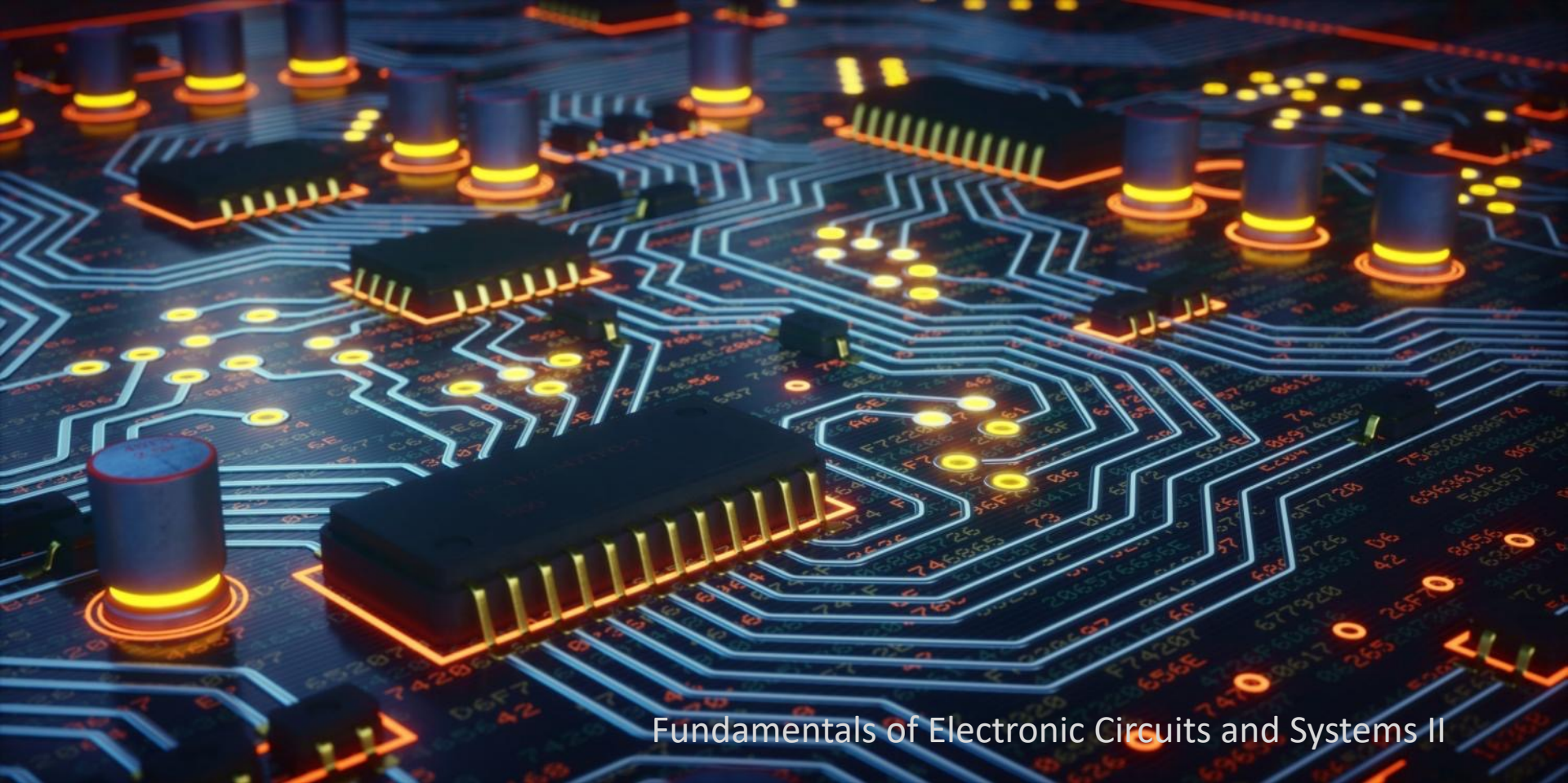




Fundamentals of Electronic Circuits and Systems II

# Oscillators & Signal Generators

Milin Zhang

Dept of EE, Tsinghua University

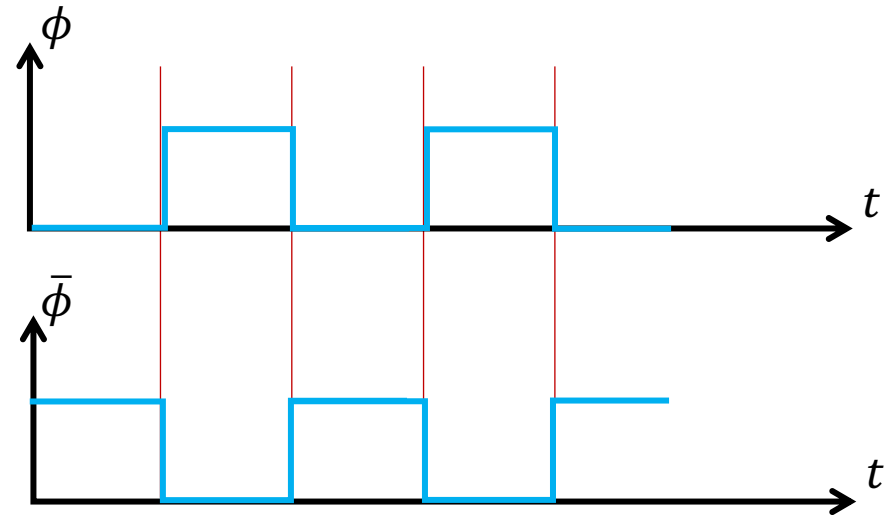
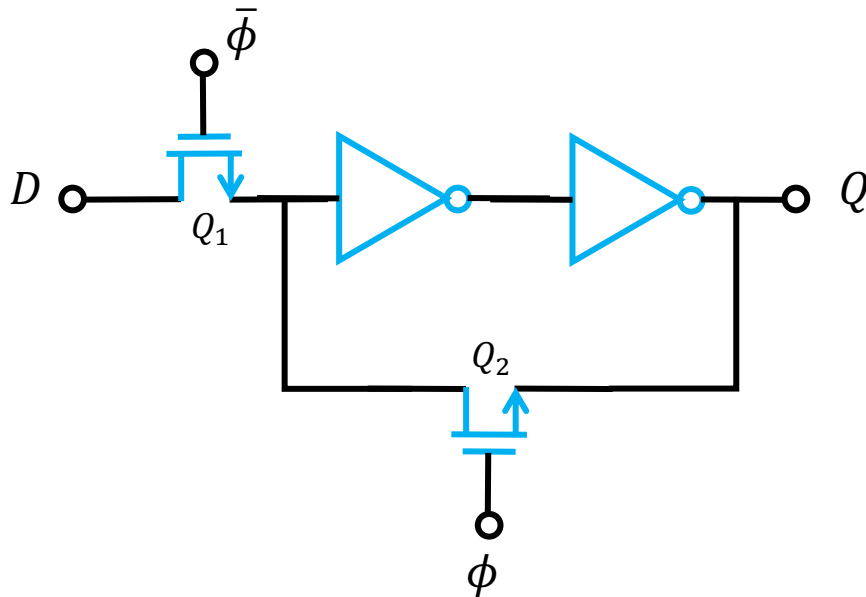


# Outline

- HOW to generate an oscillation?
- Linear Oscillator
- Non-linear Oscillator

# Recall: Dynamic Circuit

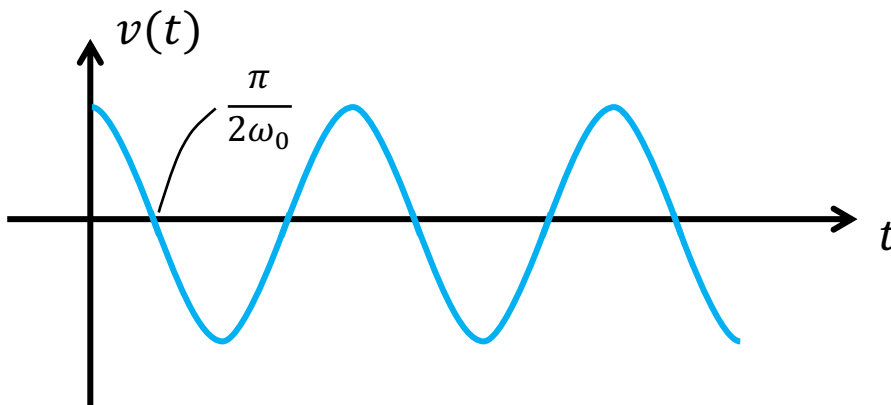
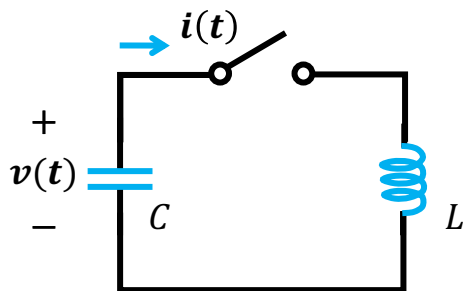
**QUESTION:** Find out the output  $Q$  with different input  $D$ .  $\phi$  is a clock signal. The input switches between  $V_{dd}$  and  $GND$



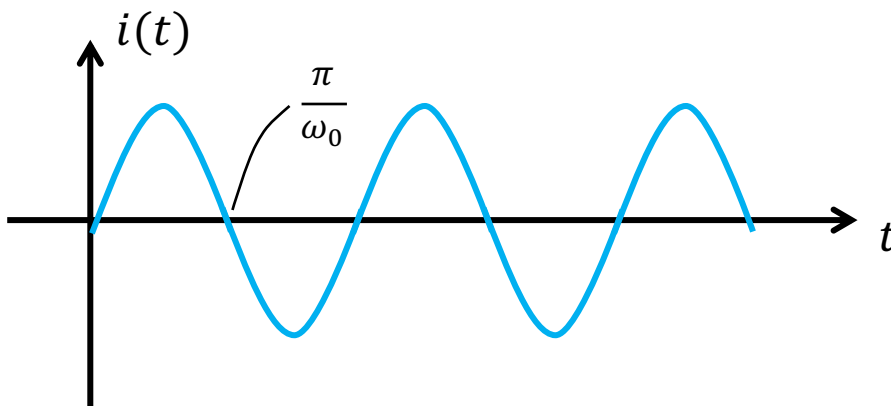
**HOW to generate a CLOCK?**

# Recall: Source free LC circuit

**QUESTION:** Find how the voltage on the capacitor and the current through the inductor change after the switch is turned on.

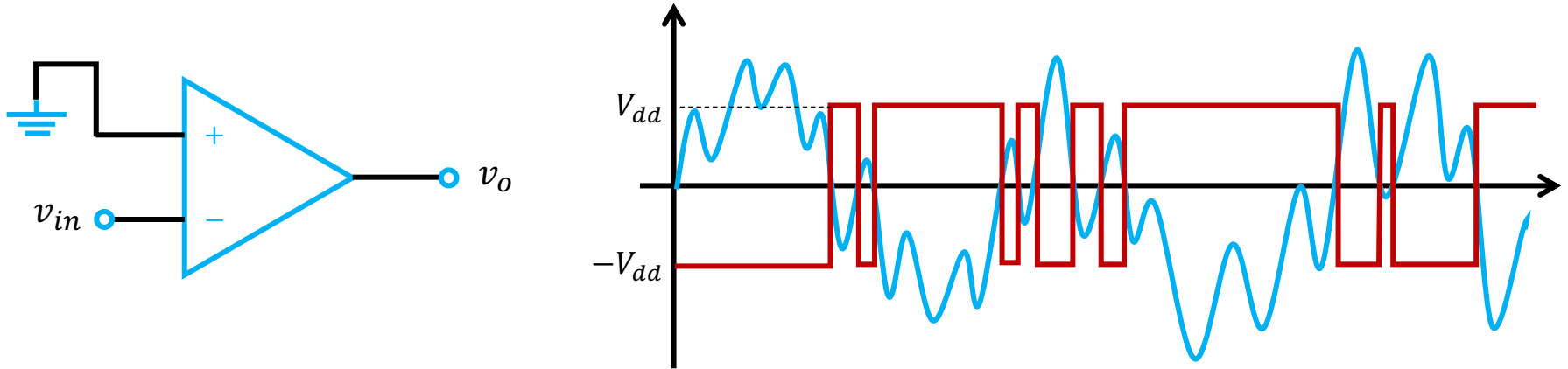


$$\begin{cases} v(t) = A\cos(\omega_0 t + \theta) \\ i(t) = \sqrt{\frac{C}{L}} A \sin(\omega_0 t + \theta) \end{cases}$$



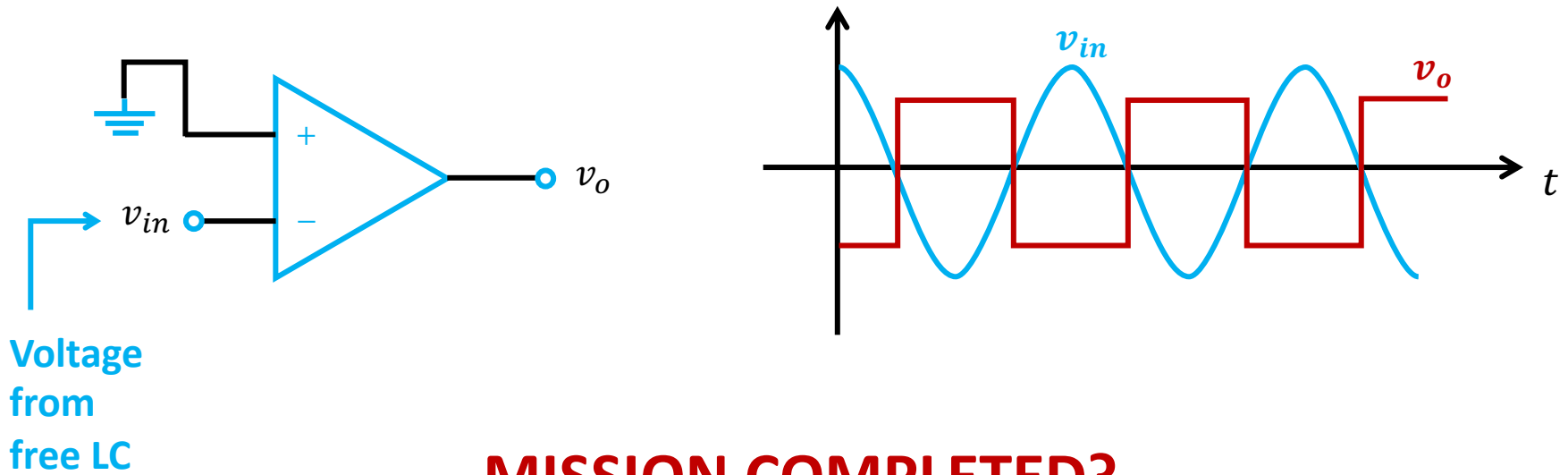
# Recall: zero-crossing detector

**QUESTION:** Find the output of the circuit. The op-amp is ideal.



- If  $v_{in} > 0 \rightarrow v_o = -V_{dd}$
- If  $v_{in} < 0 \rightarrow v_o = V_{dd}$

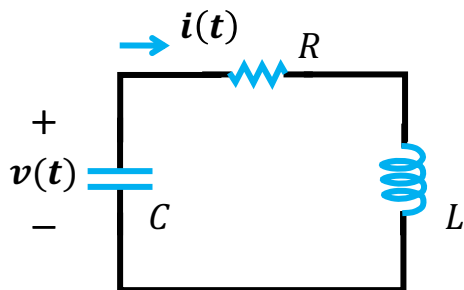
# How to generate a clock?



**MISSION COMPLETED?**

# Recall: source free RLC circuit

**QUESTION:** Find how the voltage on the capacitor and the current through the inductor change.



$$v_c(t) = e^{-\zeta\omega_0 t} [A_1 \cos(\omega_d t) + A_2 \sin(\omega_d t)]$$

Define **DAMPING FACTOR**

$$\alpha = \frac{R}{2L}$$

Define **RESONATE FREQUENCY**

$$\omega_0 = \frac{1}{\sqrt{LC}}$$

Define **DAMPING RATIO**

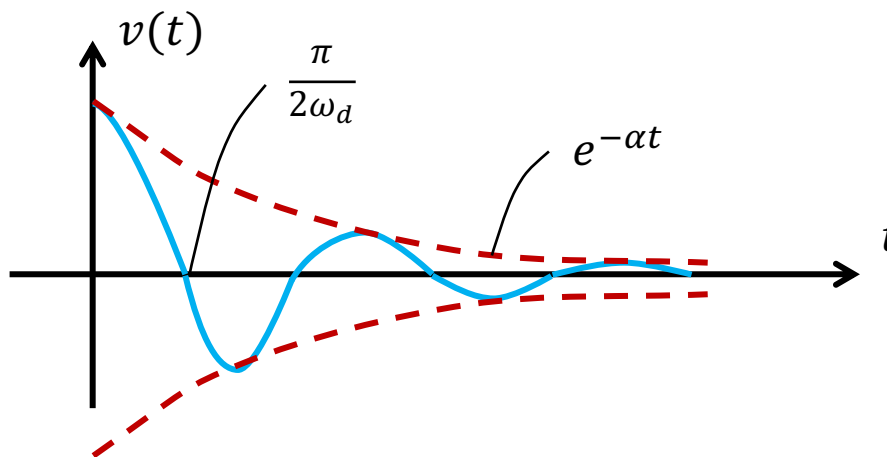
$$\zeta = \frac{\alpha}{\omega_0} = \frac{R}{2\omega_0 L}$$

$$\frac{d^2 v(t)}{dt^2} + \frac{R}{L} \frac{dv(t)}{dt} + \frac{1}{LC} v(t) = 0$$

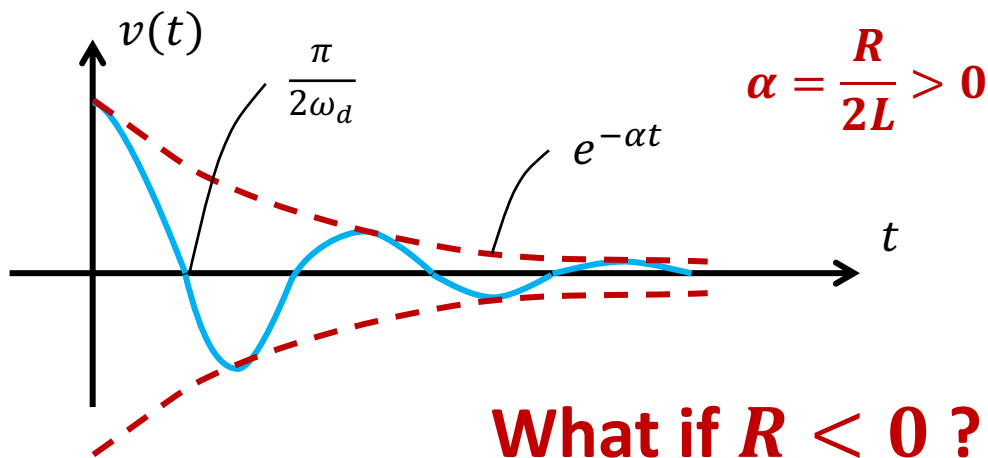
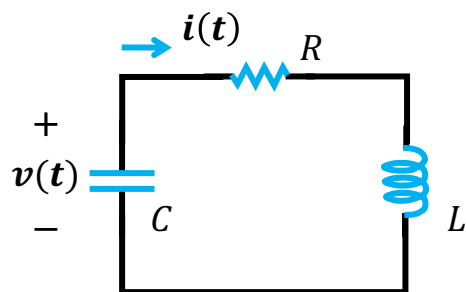
Solution:  $v(t) = K_1 e^{s_1 t} + K_2 e^{s_2 t}$

Where 
$$\begin{cases} s_1 = -\zeta\omega_0 + \omega_0\sqrt{\zeta^2 - 1} \\ s_2 = -\zeta\omega_0 - \omega_0\sqrt{\zeta^2 - 1} \end{cases}$$

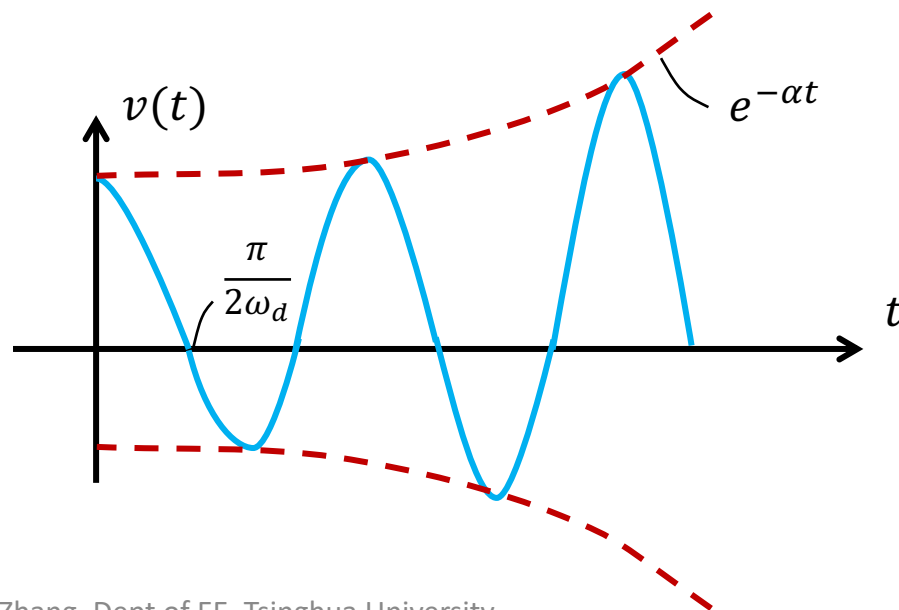
and  $\omega_0 = \frac{1}{\sqrt{LC}}, \zeta = \frac{R}{2\omega_0 L}$



# Source free RLC circuit

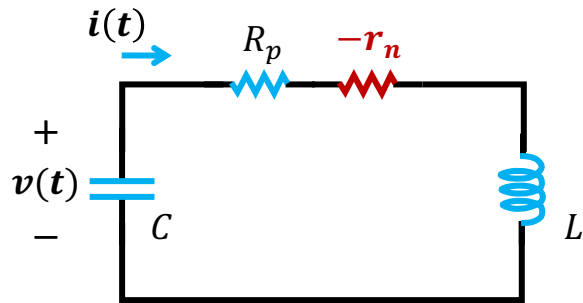


$$\left[ \begin{aligned} v_c(t) &= K_1 e^{-\alpha t} \cos(\omega_d t + \theta_1) \\ \left\{ \begin{aligned} K_1 &= \sqrt{v_{C_1}^2(0) + \left( \frac{\alpha C_1 v_{C_1}(0) - i_{L_1}(0)}{C_1 \omega_d} \right)^2} \\ \theta_1 &= \tan^{-1} \left( \frac{\alpha C_1 v_{C_1}(0) - i_{L_1}(0)}{C_1 \omega_d v_{C_1}(0)} \right) \end{aligned} \right. \end{aligned} \right]$$





# Negative Resistance

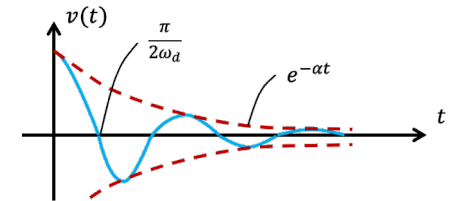


$$\left[ \begin{array}{l} K_1 = \sqrt{v_{C_1}^2(0) + \left( \frac{\alpha C_1 v_{C_1}(0) - i_{L_1}(0)}{C_1 \omega_d} \right)^2} \\ \theta_1 = \tan^{-1} \left( \frac{\alpha C_1 v_{C_1}(0) - i_{L_1}(0)}{C_1 \omega_d v_{C_1}(0)} \right) \end{array} \right]$$

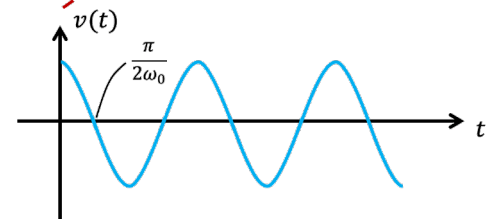
$$v_c(t) = K_1 e^{-\alpha t} \cos(\omega_d t + \theta_1)$$

Where  $\alpha = \frac{R_p - r_n}{2L}$

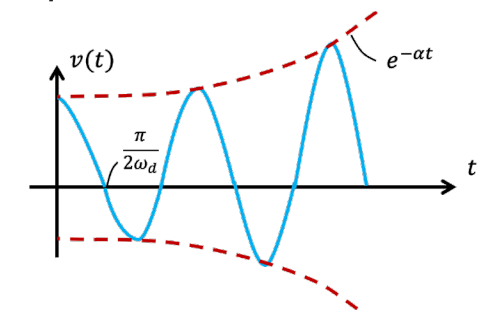
$R_p > r_n$



$R_p = r_n$



$R_p < r_n$

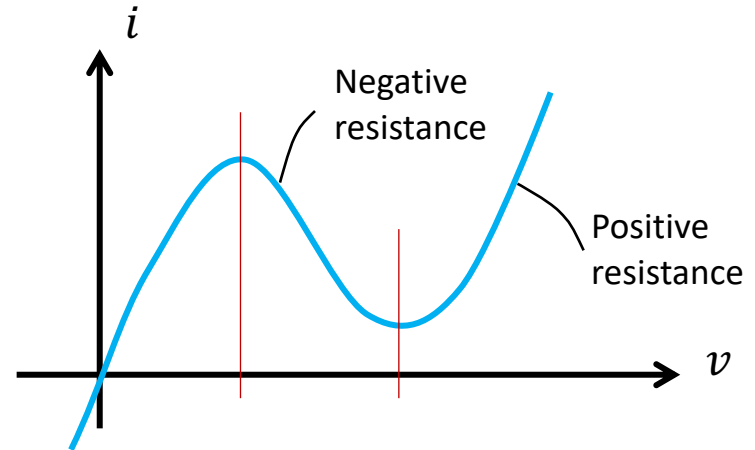


# Devices with negative resistance

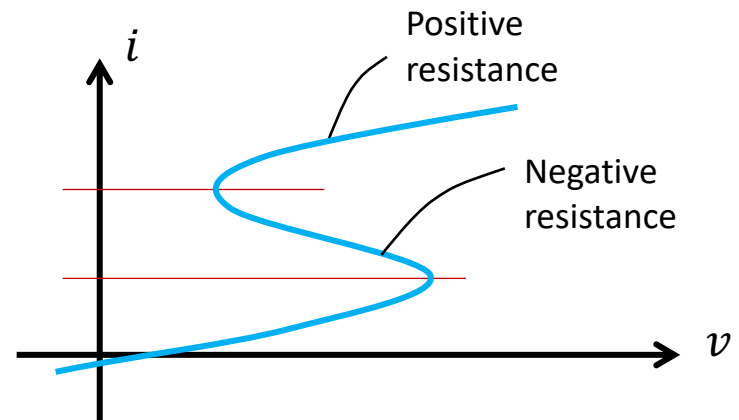
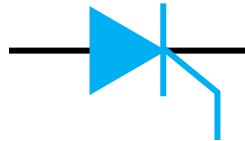
- Tunnel Diode



- Gunn Diode

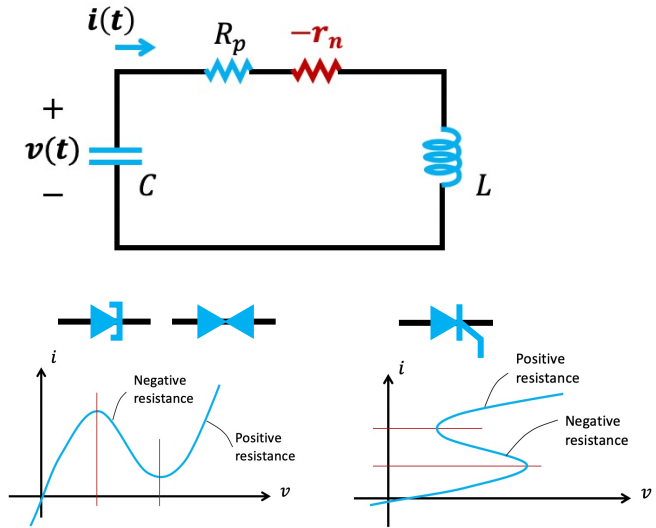


- Thyristor



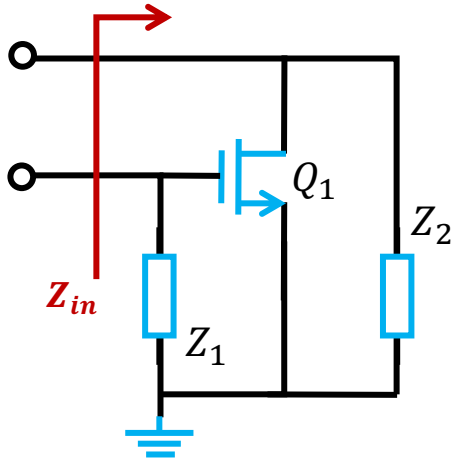
# Outline

- HOW to generate an oscillation?
  - Negative resistance
  - Devices with negative resistance
  - **Circuit features negative resistance**



# Example 1

**QUESTION:** Calculate the equivalent impedance  $Z_{in}$  of the circuit.



- **Step 1: perform DC analysis**

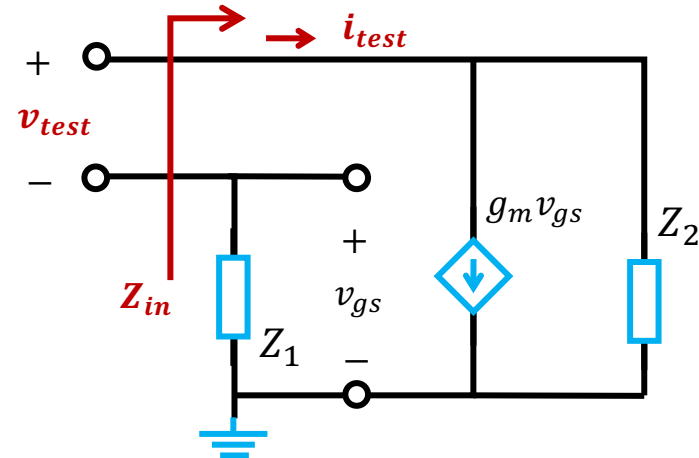
Assume  $Q_1$  is biased in saturation region according to DC analysis

- **Step 2: perform AC analysis**

- **Step 2.1: replace the transistor with the small-signal model**
- **Step 2.2: turn off DC sources**
  - SHORT all voltage sources
  - OPEN all current sources
- **Step 2.3: Calculate small-signal model parameters**

# Example 1

**QUESTION:** Calculate the equivalent impedance  $Z_{in}$  of the circuit.



- According to KCL

$$i_{test} = g_m v_{gs} + i_{Z_2}$$

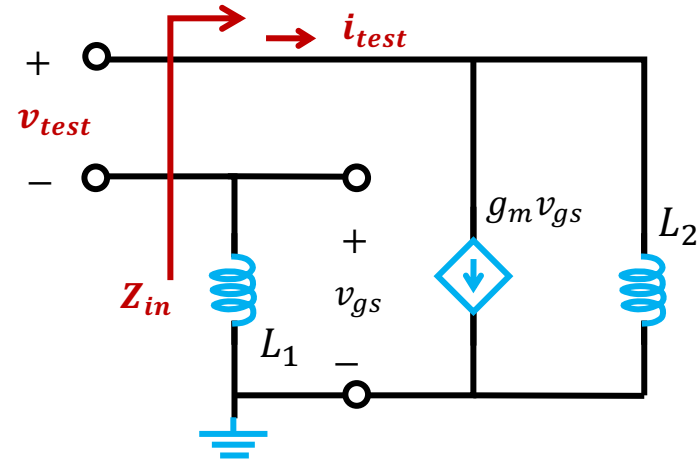
$$= g_m v_{gs} + \frac{v_{test} + v_{gs}}{Z_2}$$

$$= -g_m Z_1 i_{test} + \frac{v_{test} - Z_1 i_{test}}{Z_2}$$

$$\Rightarrow Z_{in} = \frac{v_{test}}{i_{test}} = Z_1 + Z_2 + g_m Z_1 Z_2$$

# Example 1

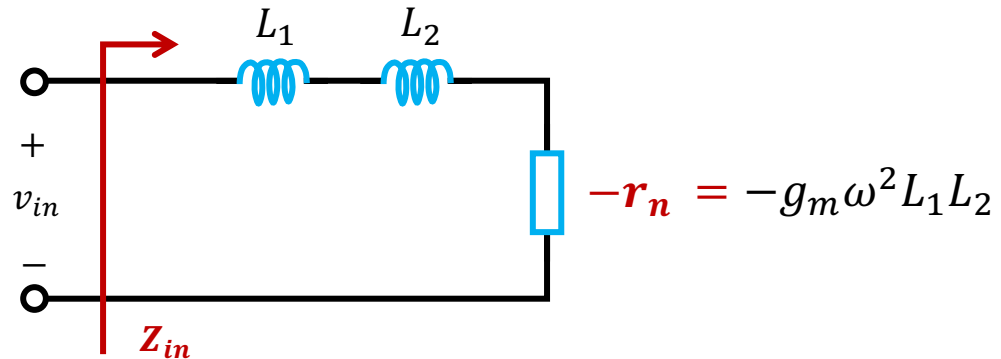
**QUESTION:** Calculate the equivalent impedance  $Z_{in}$  of the circuit.



$$Z_{in} = \frac{v_{test}}{i_{test}} = Z_1 + Z_2 + g_m Z_1 Z_2$$

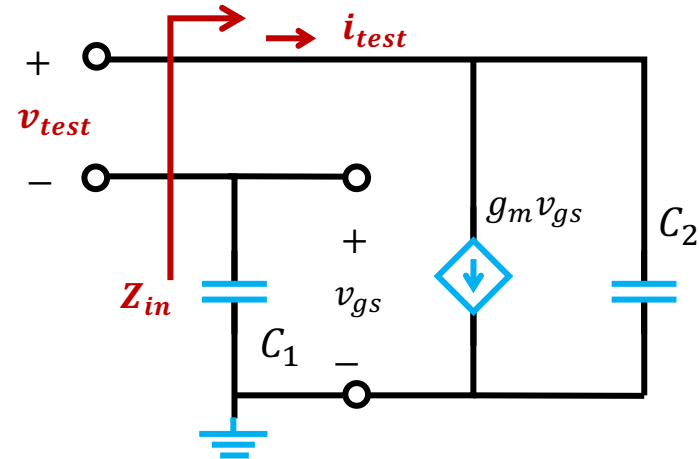
- If  $Z_1$  and  $Z_2$  are inductors

$$\begin{aligned} Z_{in} &= \frac{v_{test}}{i_{test}} = Z_1 + Z_2 + g_m Z_1 Z_2 \\ &= j\omega L_1 + j\omega L_2 - g_m \omega^2 L_1 L_2 \end{aligned}$$



# Example 1

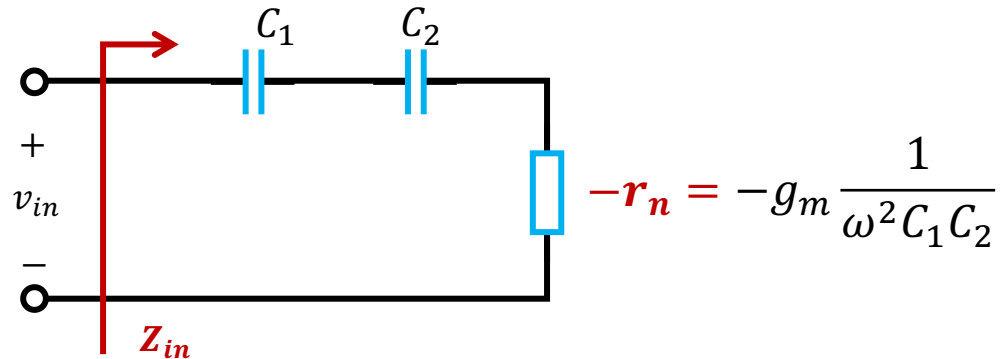
**QUESTION:** Calculate the equivalent impedance  $Z_{in}$  of the circuit.



$$Z_{in} = \frac{v_{test}}{i_{test}} = Z_1 + Z_2 + g_m Z_1 Z_2$$

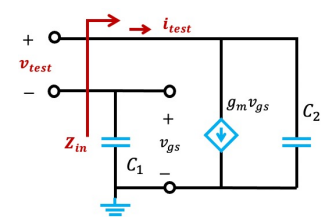
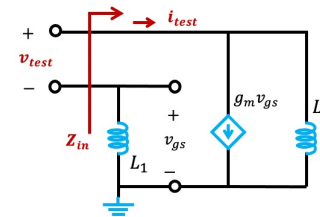
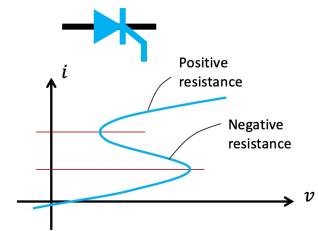
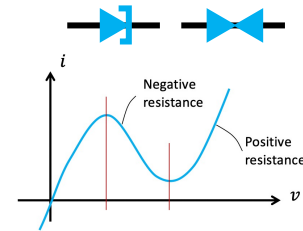
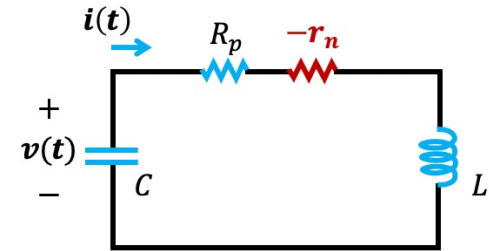
- If  $Z_1$  and  $Z_2$  are capacitors

$$\begin{aligned} Z_{in} &= \frac{v_{test}}{i_{test}} = Z_1 + Z_2 + g_m Z_1 Z_2 \\ &= \frac{1}{j\omega C_1} + \frac{1}{j\omega C_2} - g_m \frac{1}{\omega^2 C_1 C_2} \end{aligned}$$



# Outline

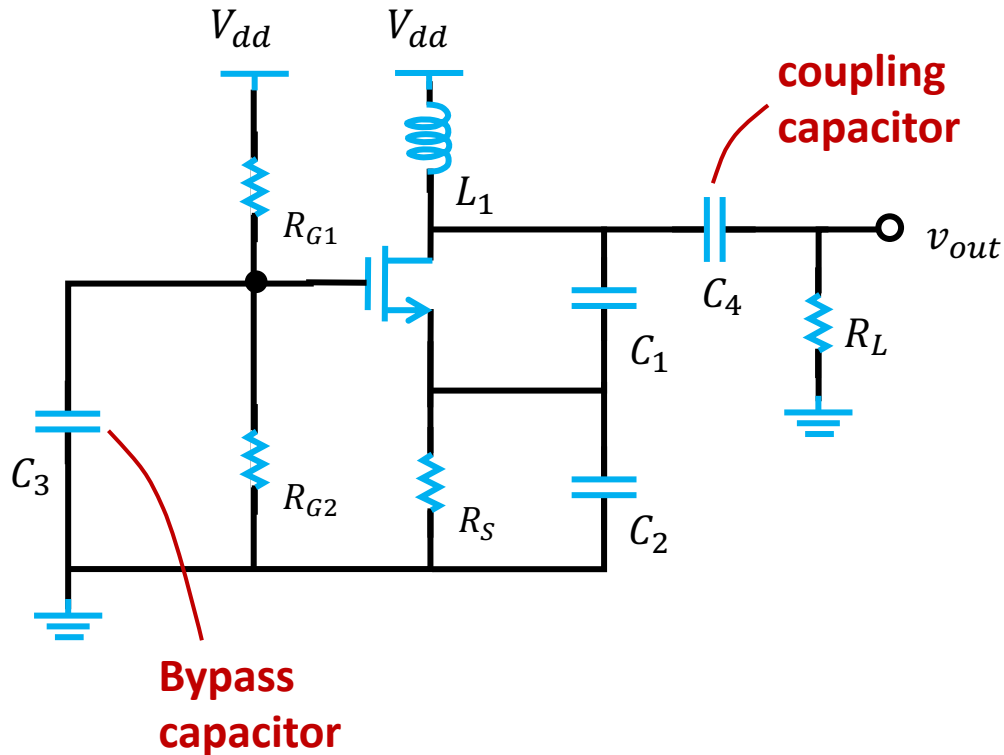
- HOW to generate an oscillation?
  - Negative resistance
  - Devices with negative resistance
  - Circuit features negative resistance
- Linear Oscillator
  - LC Oscillator





# Example 2: Colpitts Oscillator

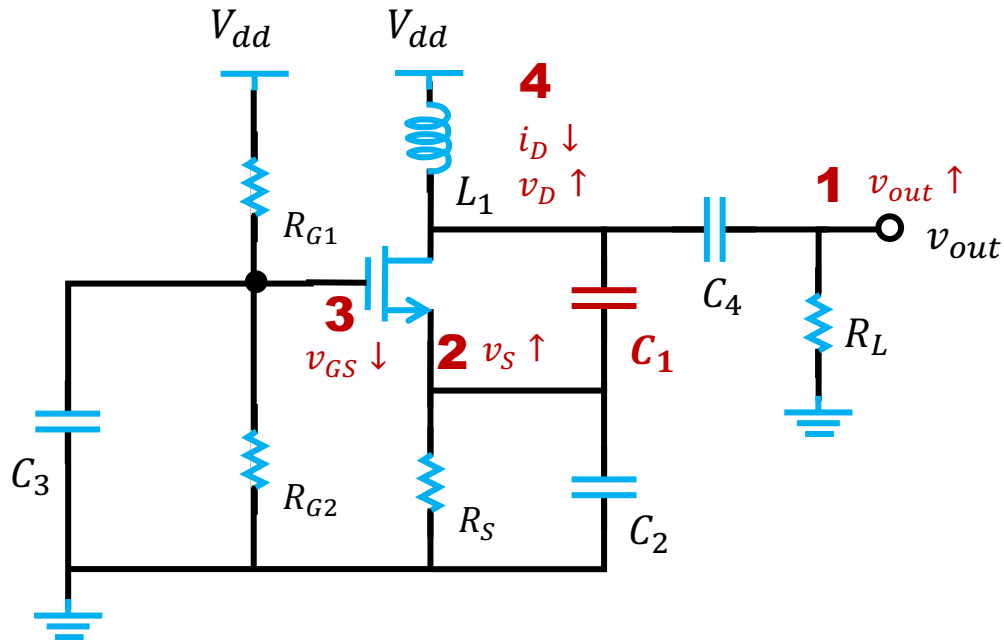
**QUESTION:** Find the functionality of the following circuit.



- $C_4$  is coupling capacitors
- $C_3$  is bypass capacitor
- $C_3$  and  $C_4$  IDEALLY are short circuit @AC

# Example 2: Colpitts Oscillator

**QUESTION:** Find the functionality of the following circuit.



**What is  $C_1$  used for?**

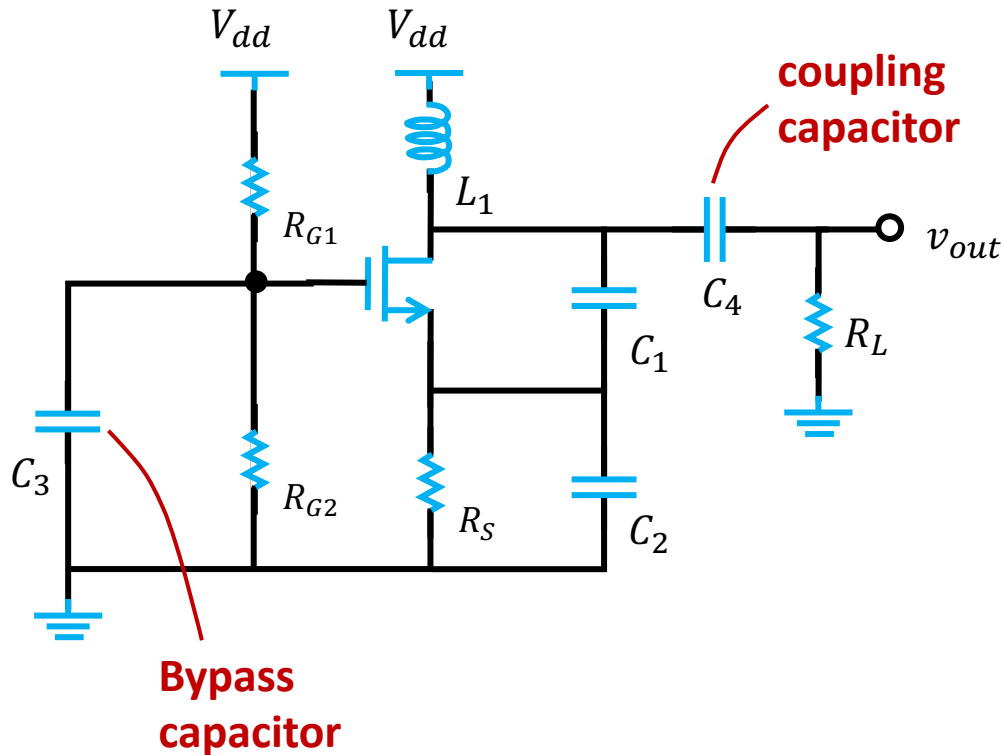
- If there is an increase @  $v_{out}$
- The voltage increase @ source
- $v_{GS}$  decrease correspondingly
- $i_D$  decreases, causing an decreasing of  $v_D$



**POSITIVE FEEDBACK is observed**

# Example 2: Colpitts Oscillator

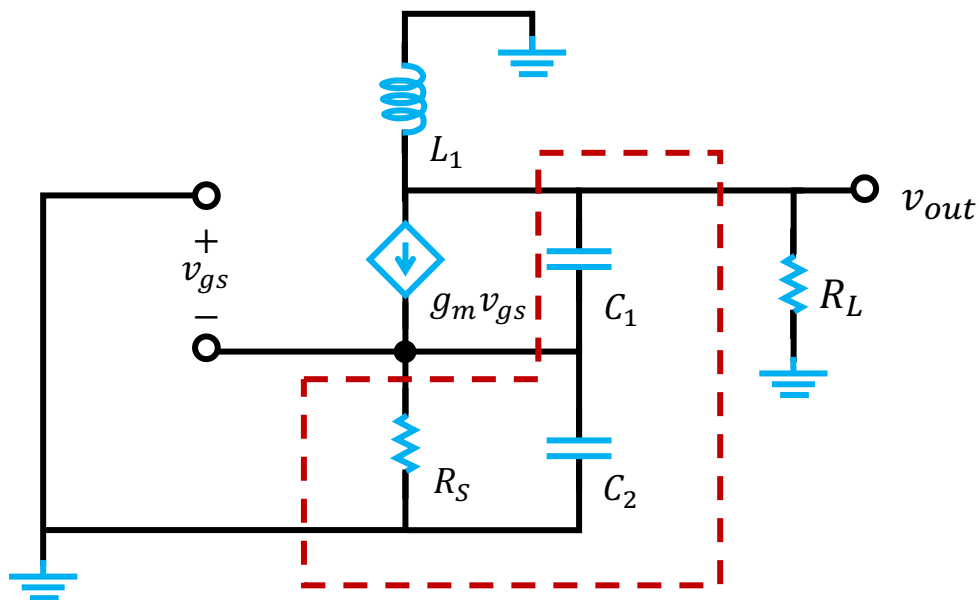
**QUESTION:** Find the functionality of the following circuit.



- **Step 1: perform DC analysis**
- **Step 2: perform AC analysis**
  - ❑ Step 2.1: turn off DC sources
  - ❑ Step 2.2: Calculate small-signal model parameters,  $g_m$
  - ❑ Step 2.3: replace the transistor with the small-signal model
  - ❑ Step 2.4: Analyze the resulting circuit

# Example 2: Colpitts Oscillator

**QUESTION:** Find the functionality of the following circuit.



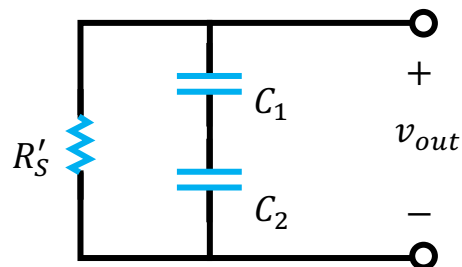
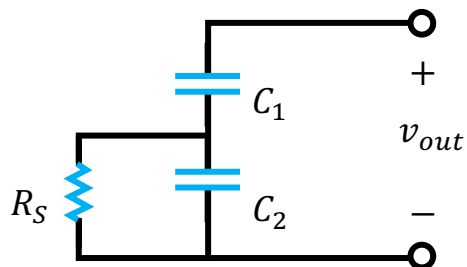
$$v_{R_S} = \frac{\frac{1}{sC_2} \parallel R_S}{\frac{1}{sC_1} + \left( \frac{1}{sC_2} \parallel R_S \right)} v_{out}$$

If  $R_S \gg \frac{1}{sC_2}$

$$\approx \frac{\frac{1}{sC_2}}{\frac{1}{sC_1} + \frac{1}{sC_2}} v_{out} = \frac{C_1}{C_1 + C_2} v_{out}$$

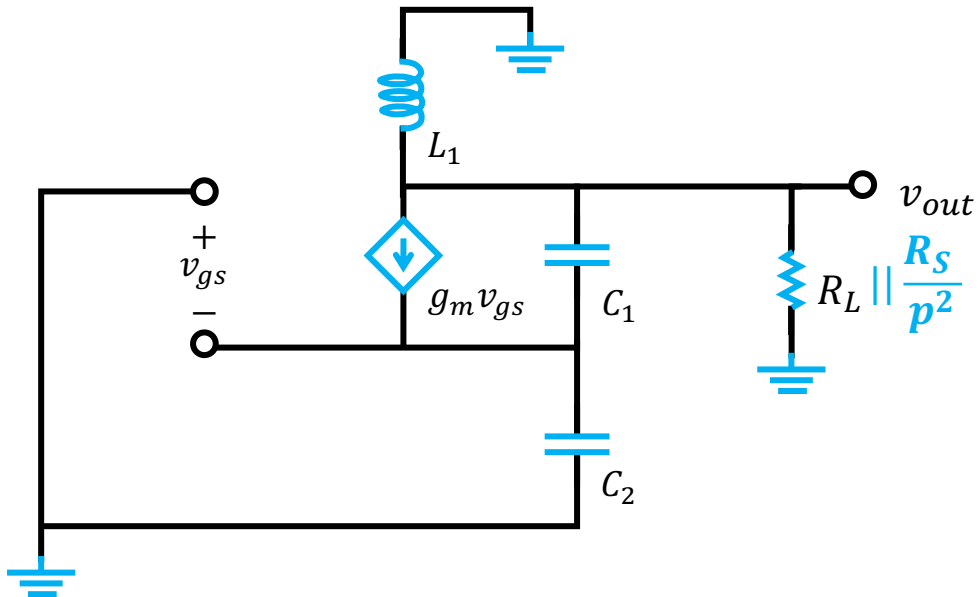
$p$

$$R'_S = \frac{R_S}{p^2} \text{ to keep a same power}$$



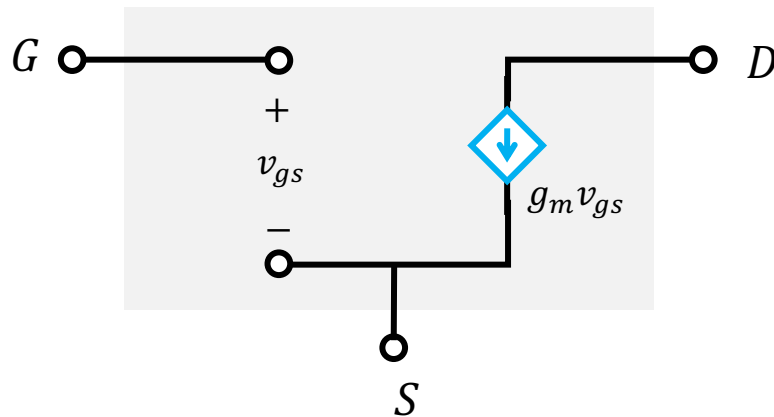
# Example 2: Colpitts Oscillator

**QUESTION:** Find the functionality of the following circuit.

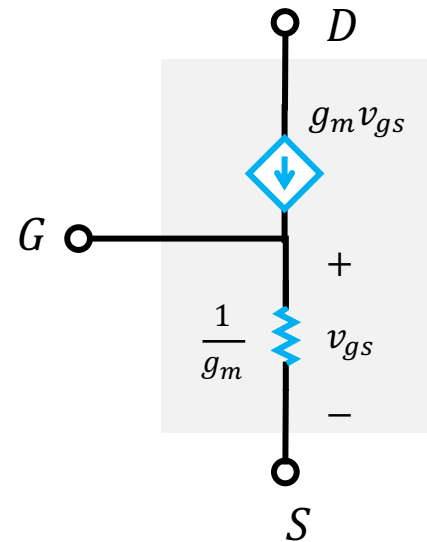


# Recall: MOSFET Small-Signal Model

**SIMPLIFIED HYBRID- $\pi$  MODEL**



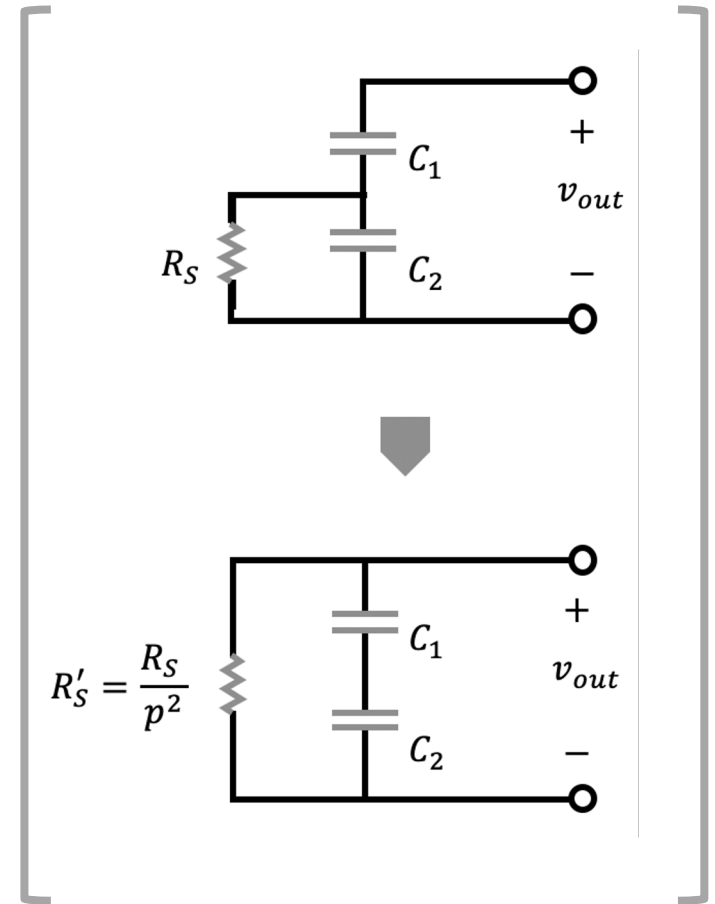
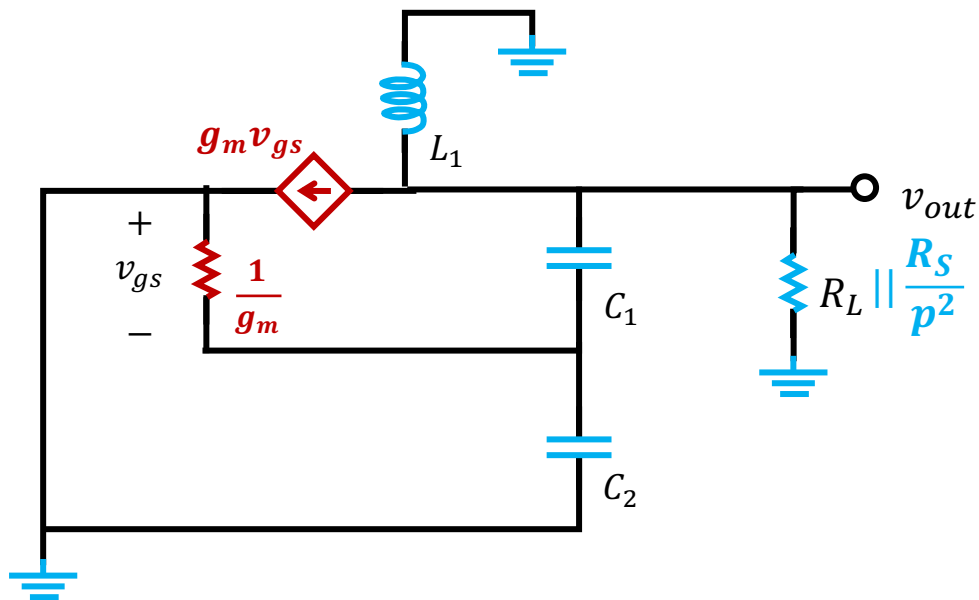
**SIMPLIFIED  $T$  MODEL**



**HYBRID- $\pi$  MODEL and  $T$  MODEL are equivalent**

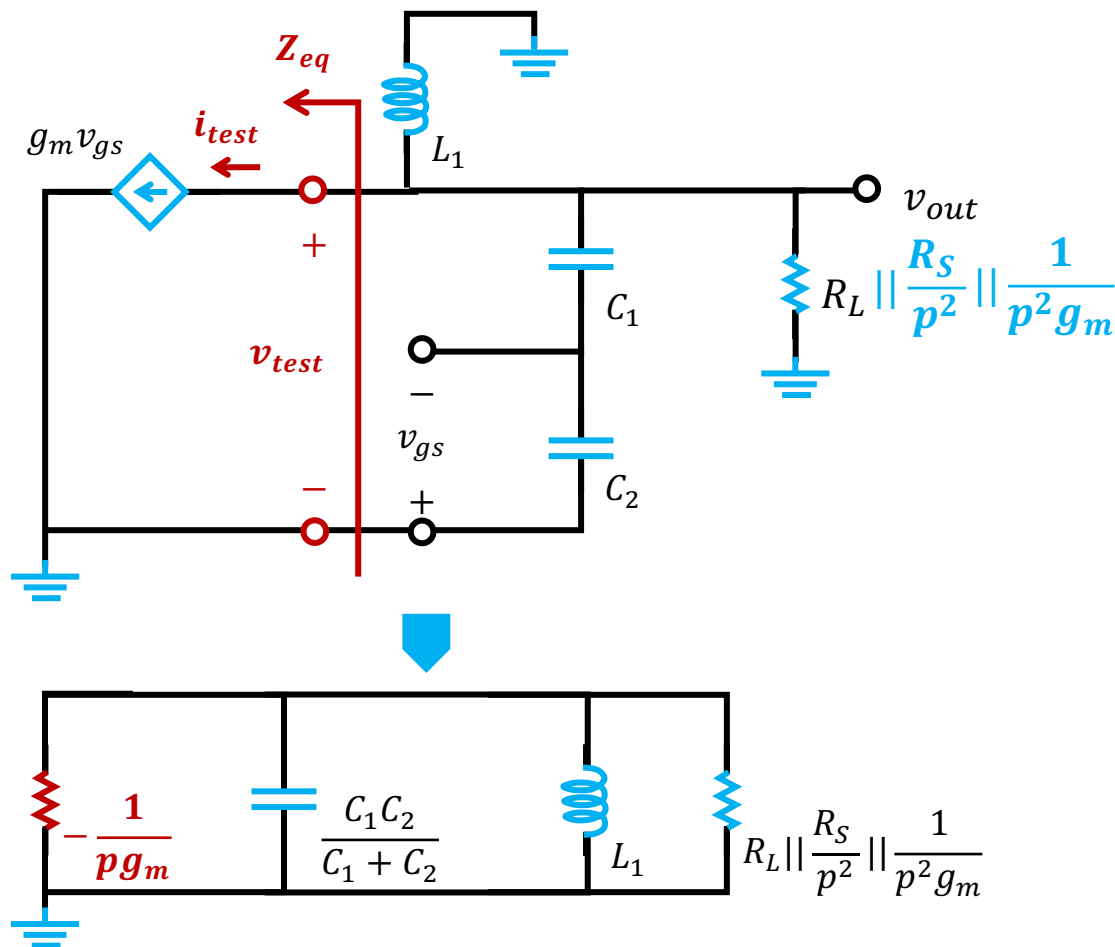
# Example 2: Colpitts Oscillator

**QUESTION:** Find the functionality of the following circuit.



# Example 2: Colpitts Oscillator

**QUESTION:** Find the functionality of the following circuit.



- According to KVL

$$\begin{aligned} v_{test} &= -sC_2 v_{gs} \frac{C_1 + C_2}{sC_1 C_2} \\ &= -\frac{C_1 + C_2}{C_1} v_{gs} \end{aligned}$$

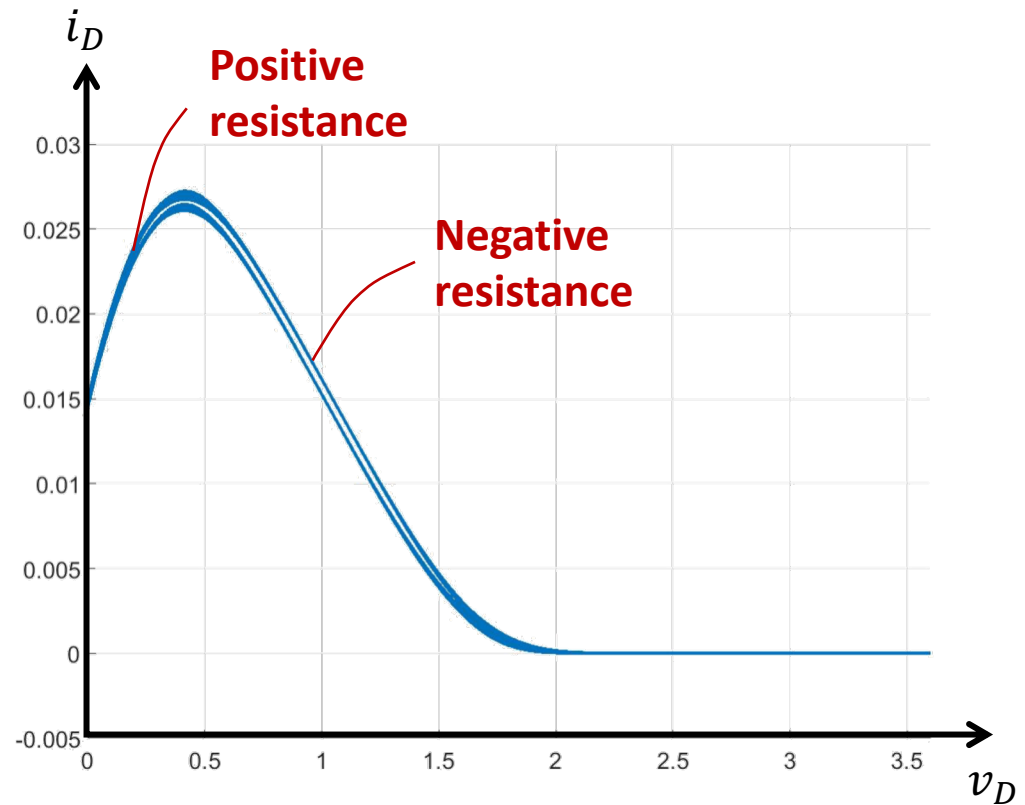
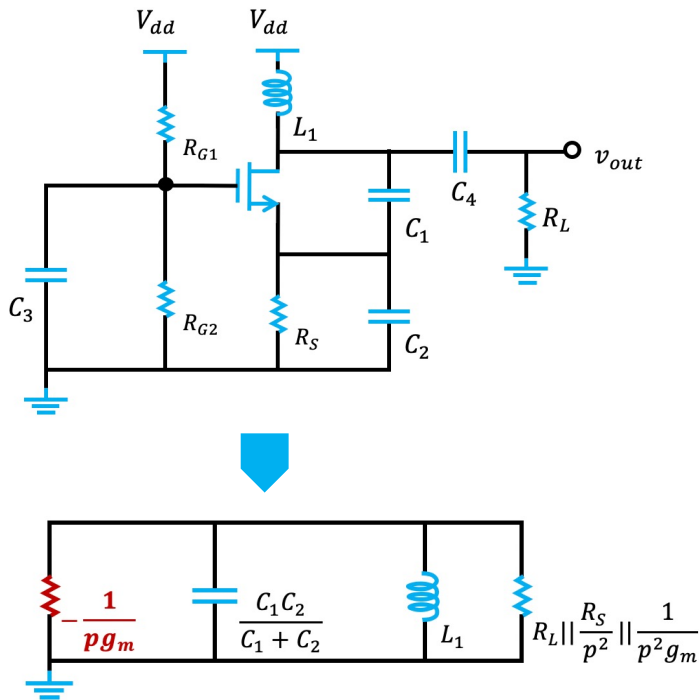
$\frac{1}{p}$

$$Z_{eq} = \frac{v_{test}}{i_{test}} = -\frac{1}{pg_m}$$



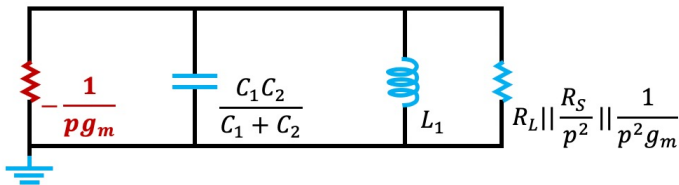
# Example 2: Colpitts Oscillator

**QUESTION:** Find the functionality of the following circuit.



# Example 2: Colpitts Oscillator

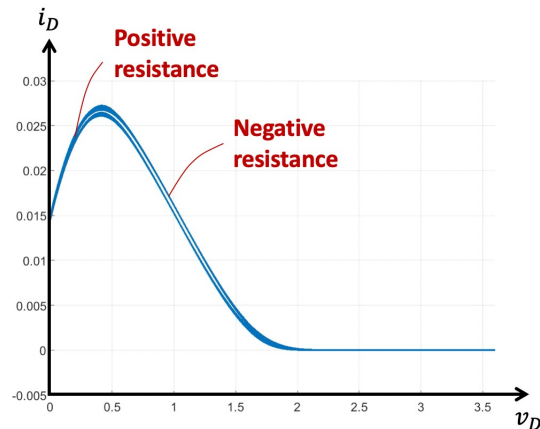
**QUESTION:** Find the functionality of the following circuit.



- Oscillation can be expected when

$$g_m p > p^2 g_m + \frac{p^2}{R_S} + \frac{1}{R_L}$$

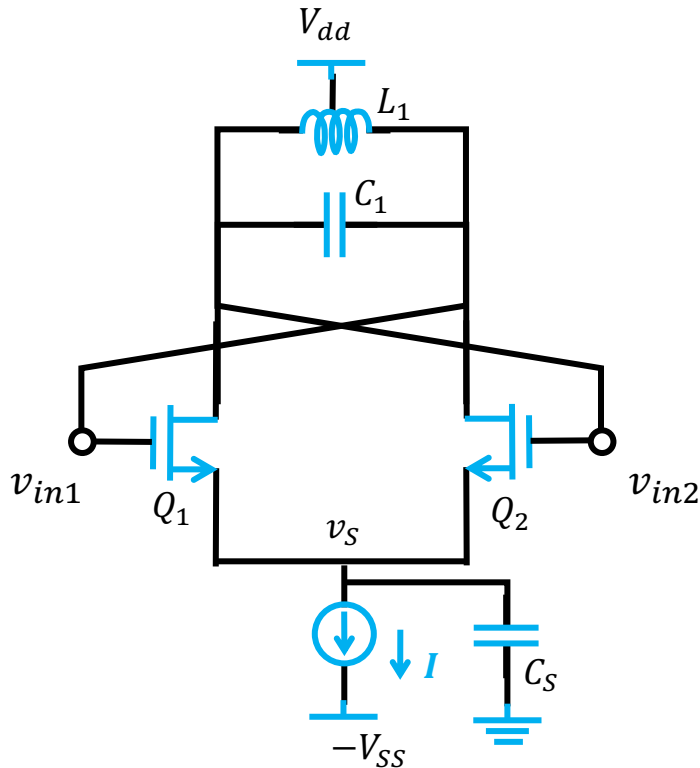
$\underline{g'_L}$



➡ 
$$g_m > \frac{g'_L}{p - p^2}$$

# Example 3

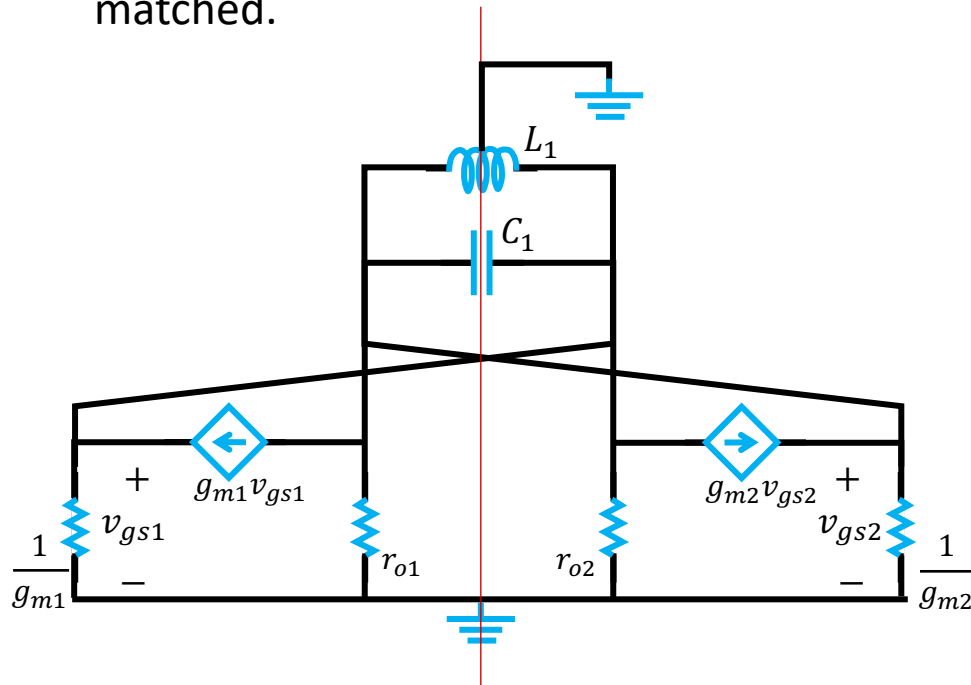
**QUESTION:** Find the functionality of the following circuit. Assume  $Q_1$  and  $Q_2$  are matched.



- **Step 1: perform DC analysis**
- **Step 2: perform AC analysis**
  - ❑ Step 2.1: turn off DC sources
  - ❑ Step 2.2: Calculate small-signal model parameters,  $g_m$
  - ❑ Step 2.3: replace the transistor with the small-signal model
  - ❑ Step 2.4: Analyze the resulting circuit

# Example 3

**QUESTION:** Find the functionality of the following circuit. Assume  $Q_1$  and  $Q_2$  are matched.



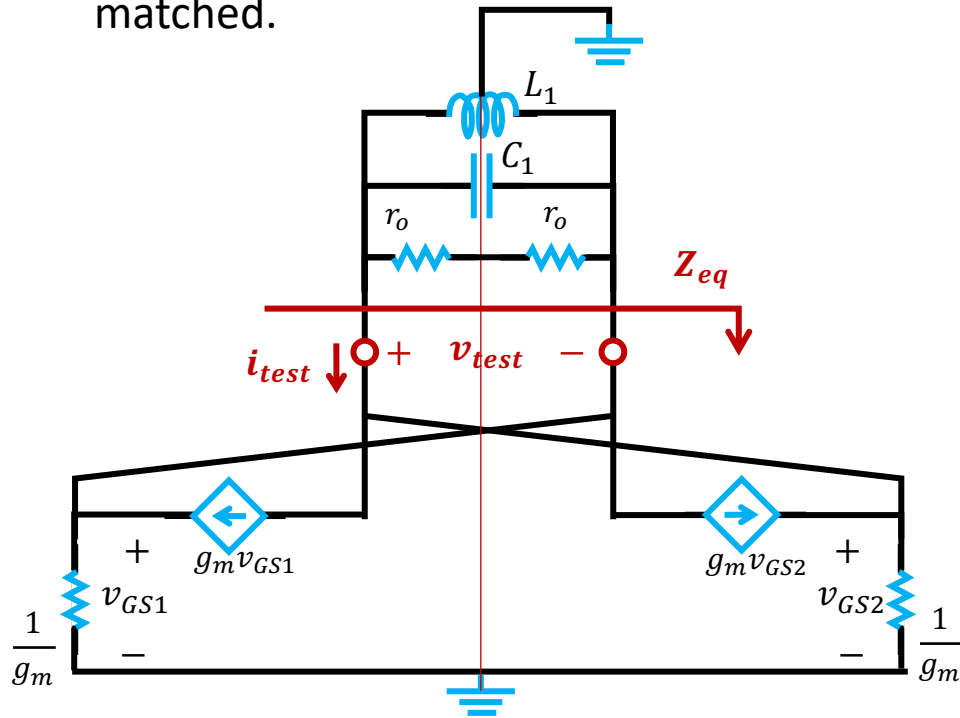
- Since  $Q_1$  and  $Q_2$  are matched

$$g_{m1} = g_{m2} = g_m$$

$$r_{o1} = r_{o2} = r_o$$

# Example 3

**QUESTION:** Find the functionality of the following circuit. Assume  $Q_1$  and  $Q_2$  are matched.



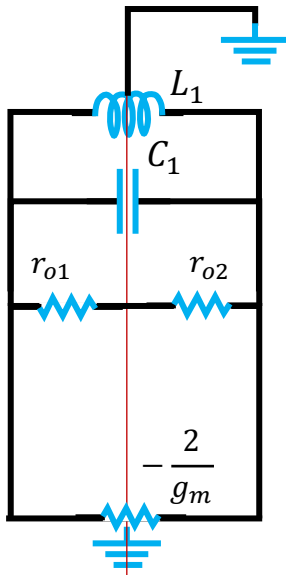
- According to KCL & resistor characteristic

$$\begin{cases} (i_{test} - g_{m1}v_{GS1} + g_{m2}v_{GS2})\frac{1}{g_{m2}} = v_{GS2} \\ (-i_{test} - g_{m2}v_{GS2} + g_{m1}v_{GS1})\frac{1}{g_{m1}} = v_{GS1} \\ v_{GS2} - v_{GS1} = v_{test} \end{cases}$$

$$\Rightarrow Z_{eq} = \frac{v_{test}}{i_{test}} = -\frac{2}{g_m}$$

# Example 3

**QUESTION:** Find the functionality of the following circuit. Assume  $Q_1$  and  $Q_2$  are matched.



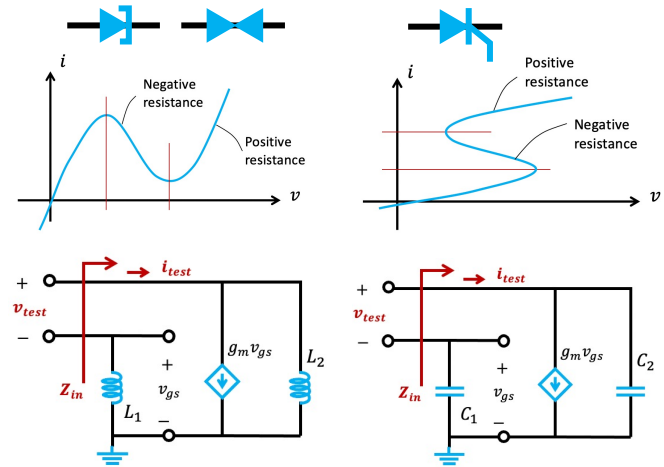
- Oscillation can be expected when

$$\frac{g_m}{2} > \frac{1}{2r_o}$$

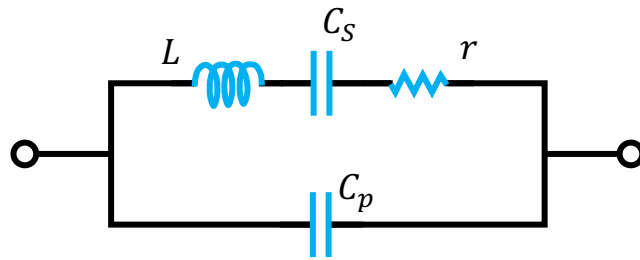
➡  $g_m > \frac{1}{r_o}$

# Outline

- HOW to generate an oscillation?
  - Negative resistance
  - Devices with negative resistance
  - Circuit features negative resistance
- Linear Oscillator
  - LC Oscillator
  - **Crystal Oscillator**

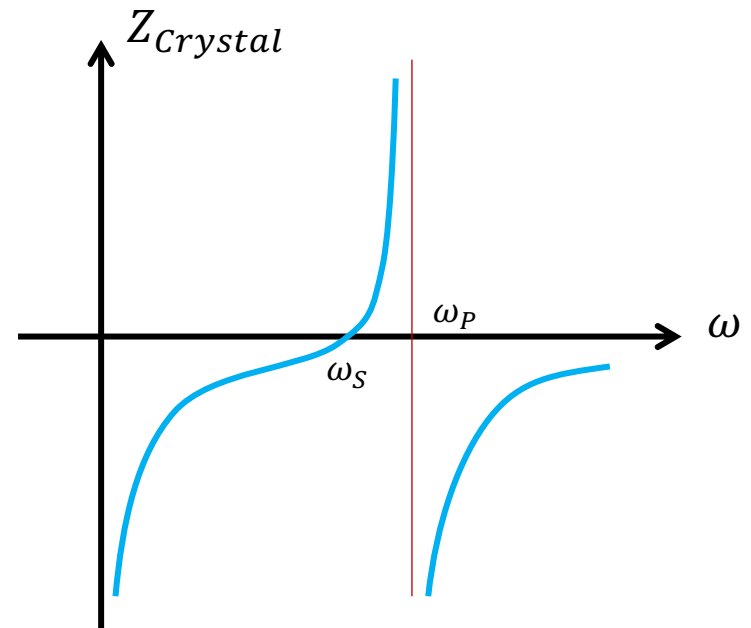


# Crystal Oscillators



Equivalent circuit

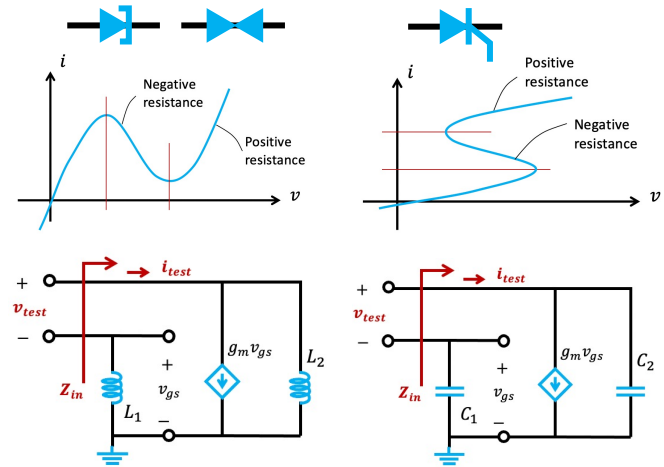
- ☺ High frequency
- ☺ High stability
- ☺ High Q factor
- ☹ Cannot integrate on chip



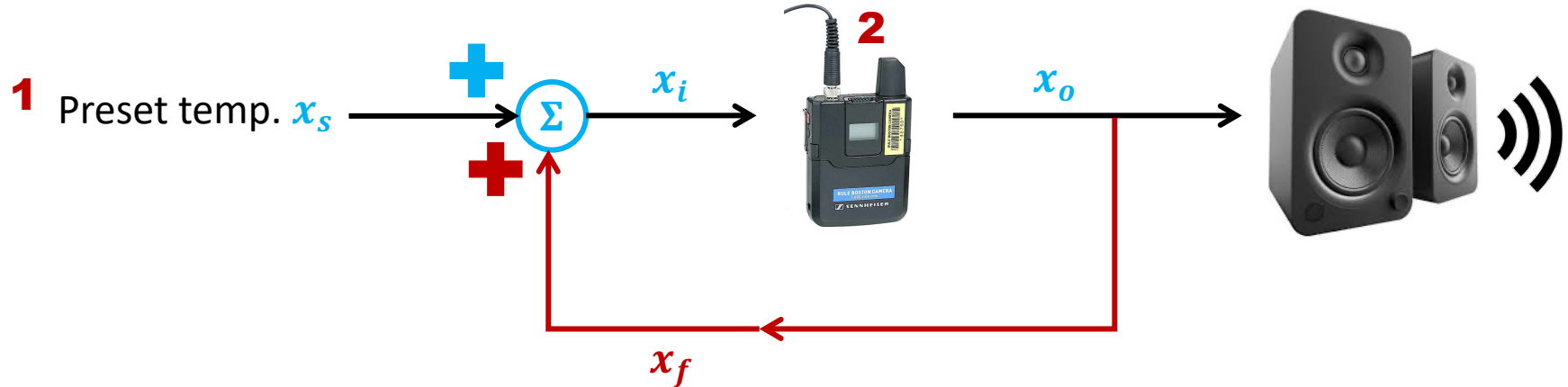


# Outline

- HOW to generate an oscillation?
  - Negative resistance
  - Devices with negative resistance
  - Circuit features negative resistance
- Linear Oscillator
  - LC Oscillator
  - Crystal Oscillator
  - **Op-Amp-RC Oscillator**



# Recall: Example of Positive Feedback



- The output  $x_o$  is related to the input  $x_i$
- The output  $x_o$  is fed to a feedback network to generate  $x_f$
- The input of the system turns into

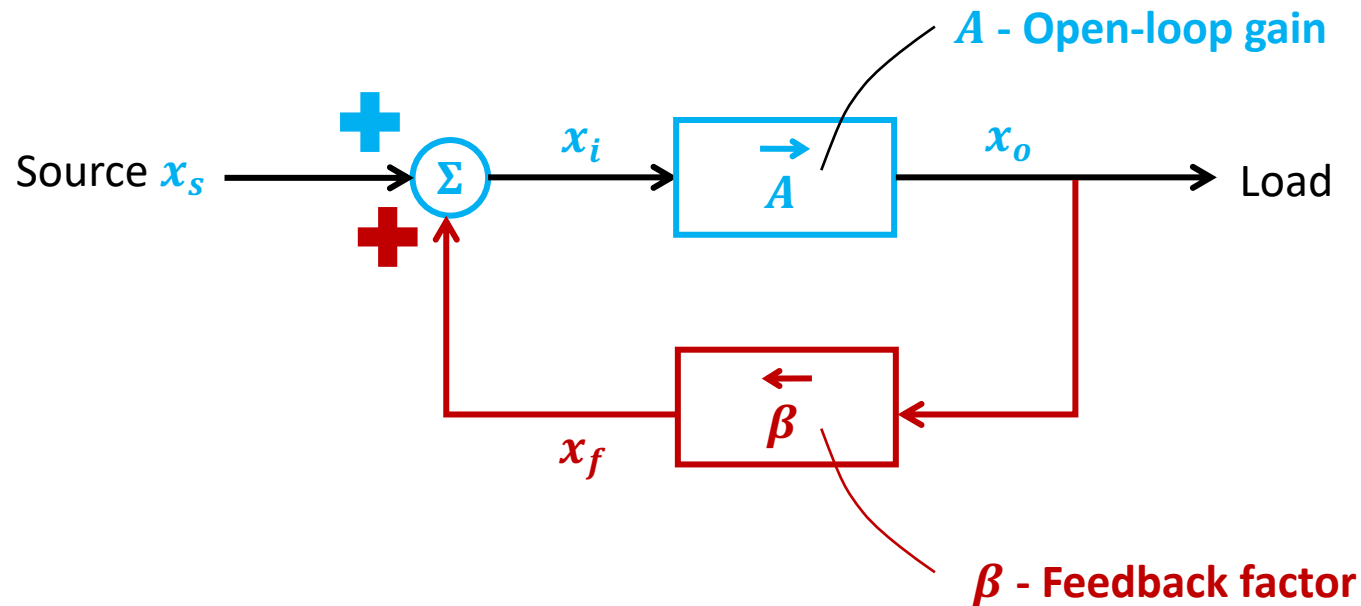
$$x_i = x_s + x_f$$



**THIS IS A POSITIVE FEEDBACK**

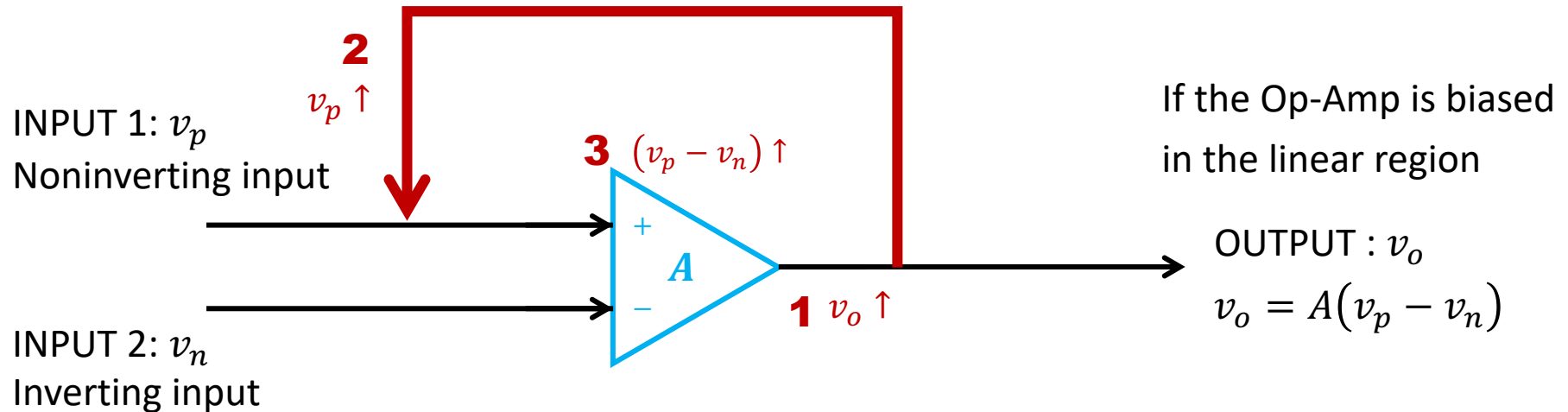
The output is fed-back to input, to  
**INCREASE** intensity of input signal

# Recall: general pos. feedback structure



- The output  $x_o$  is related to the input  $x_i$   $x_o = Ax_i$
- The output is fed to a feedback network  $x_f = \beta x_o$
- The input of the amplifier turns into  $x_i = x_s + x_f$

# Recall: Op-Amp w/ positive feedback



## How to decide if there is a positive feedback

- If there is an increase @  $v_o$   $v_o \uparrow$
- The inverting input  $v_p$  increases correspondingly
- If the op-amp is biased in the linear region,  $v_o = A(v_p - v_n)$  increases

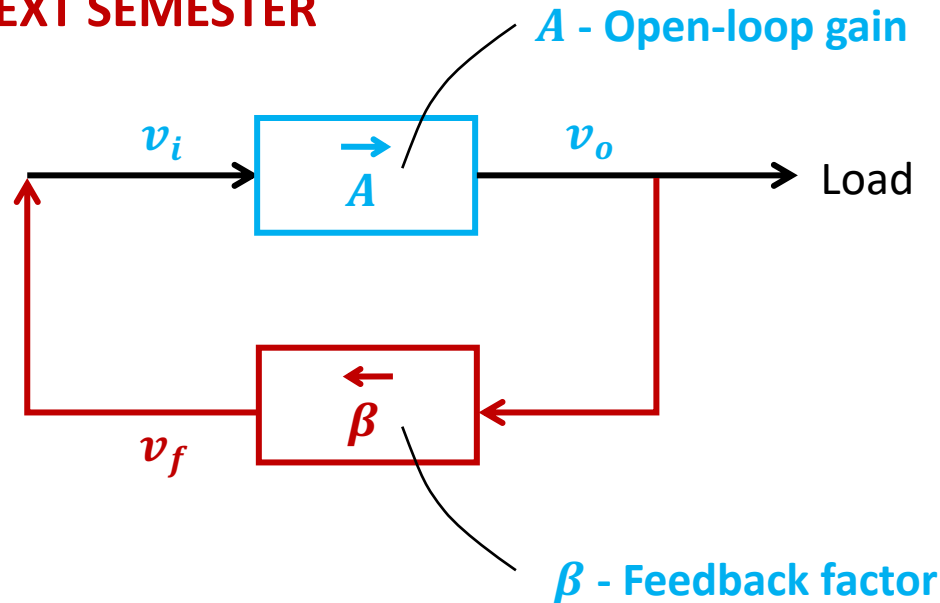
**POSITIVE FEEDBACK**

An unstable system “oscillates” between  $V_{dd}$  and  $-V_{dd}$

# Recall: positive feedback w/o input

WE WILL DISCUSS THIS NEXT SEMESTER

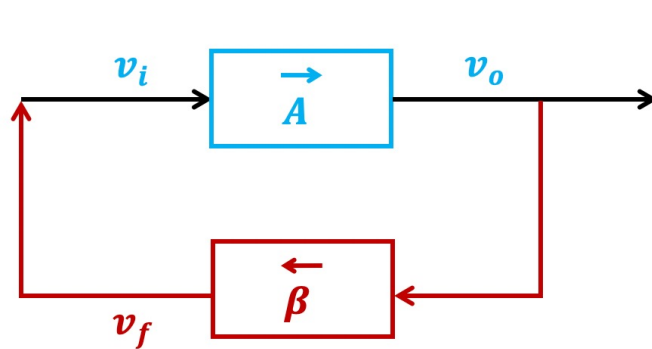
- What if there is no input in the system?
- What gonna happen if there is a disturbance?



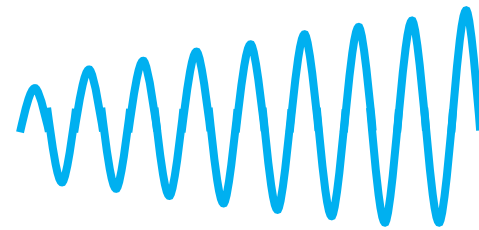
- The output  $x_o$  is related to the input  $x_i$   
$$v_o = Av_i$$
- The output is fed to a feedback network  
$$v_f = \beta v_o = A\beta v_i$$

What if  $A\beta > 1$  at the very beginning, but decreases to  $A\beta = 1$  later?

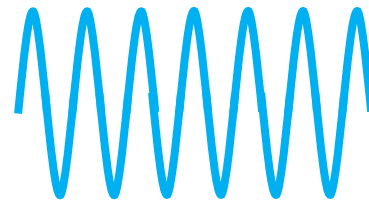
# Positive feedback w/o input



A small disturb @  $v_i$



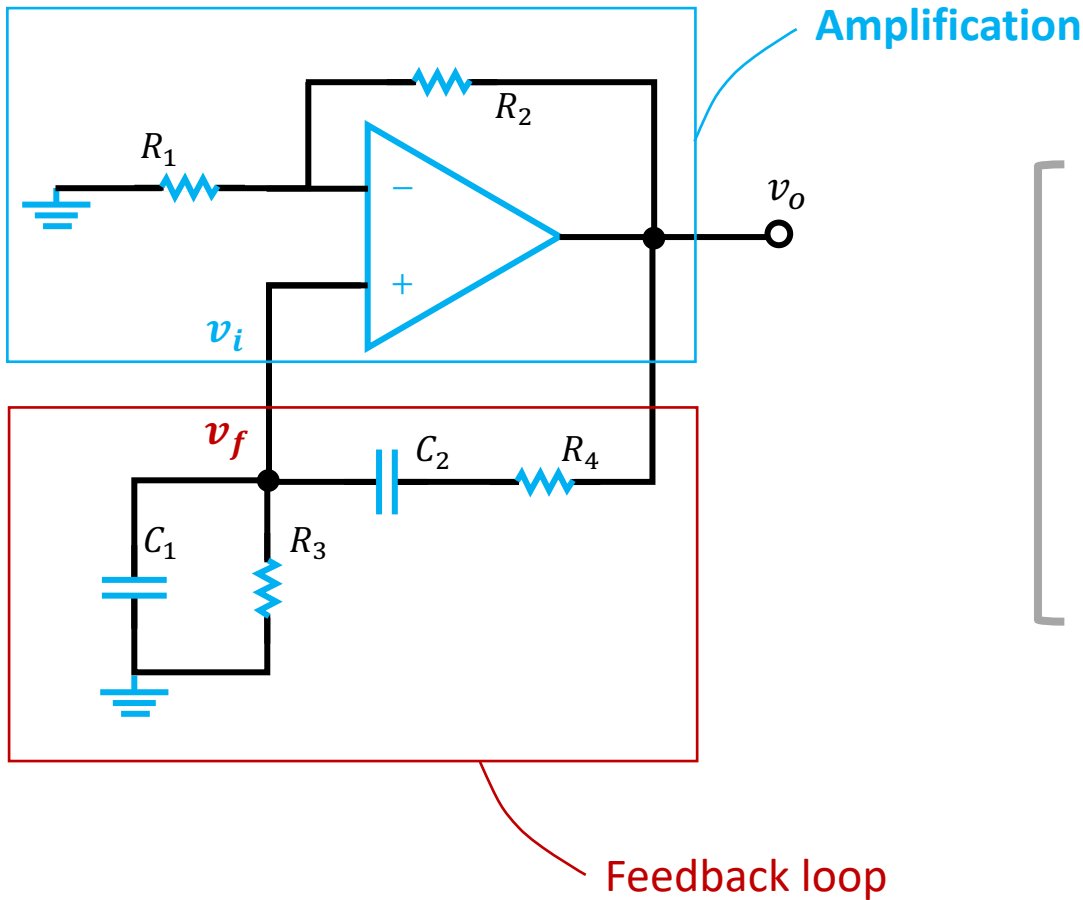
@  $A\beta > 1$   
Amplitude increases



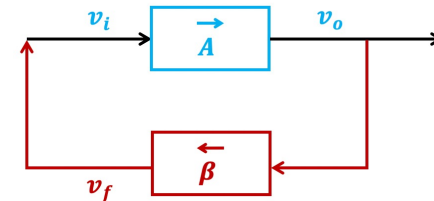
@  $A\beta = 1$   
Amplitude keeps

# Example 4: the Wien-Bridge Oscillator

**QUESTION:** Find the functionality of the following circuit with  $R_3 = R_4 = R$ ,  $C_1 = C_2 = C$ .

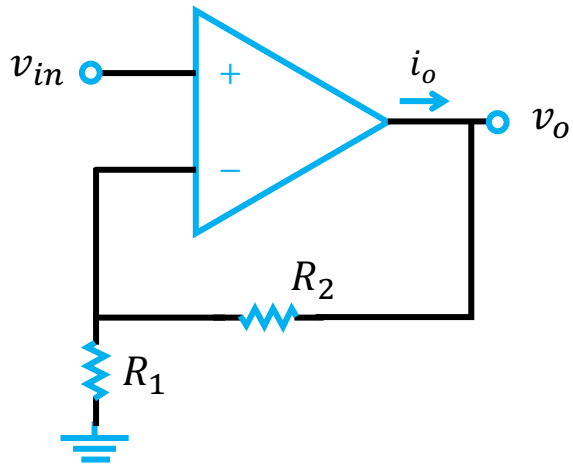


Recall: op-amp w/  
positive feedback



# Recall: The noninverting configuration

**QUESTION:** Find the functionality of the following circuit with  $R_3 = R_4 = R$ ,  $C_1 = C_2 = C$ .



$$\left\{ \begin{array}{l} i_{R_1} = \frac{v_n}{R_1} = \frac{v_{in}}{R_1} \\ i_{R_2} = \frac{v_o - v_n}{R_2} = \frac{v_o - v_{in}}{R_2} \\ i_{R_1} = i_{R_2} \end{array} \right.$$

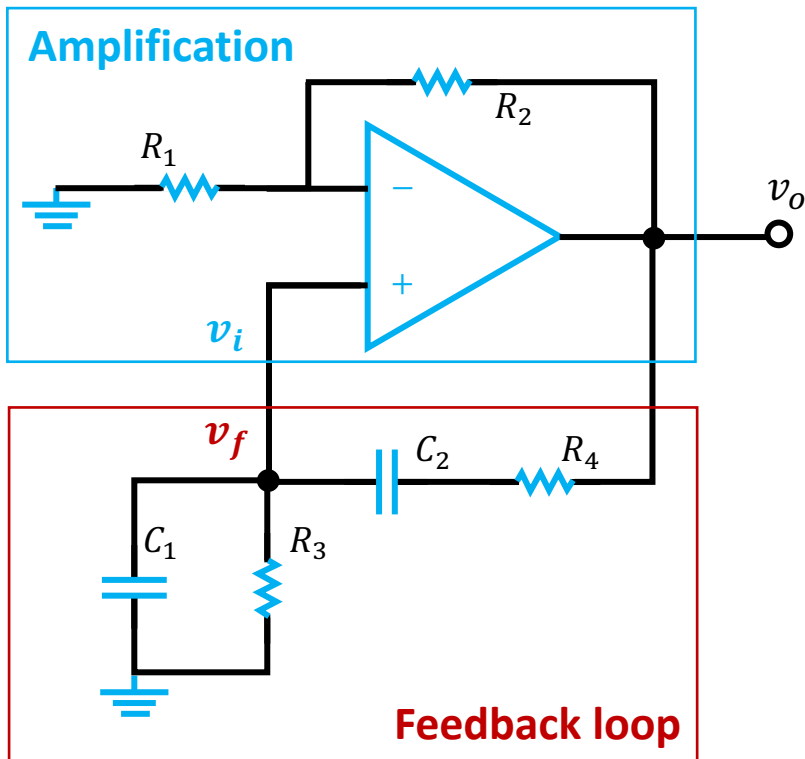
**IDEAL OP-AMP with NEGATIVE FEEDBACK enables linear region biasing**

$$\Rightarrow v_o = \frac{R_1 + R_2}{R_1} v_{in}$$



# Example 4: the Wien-Bridge Oscillator

**QUESTION:** Find the functionality of the following circuit with  $R_3 = R_4 = R$ ,  $C_1 = C_2 = C$ .



- For the amplification system, the gain

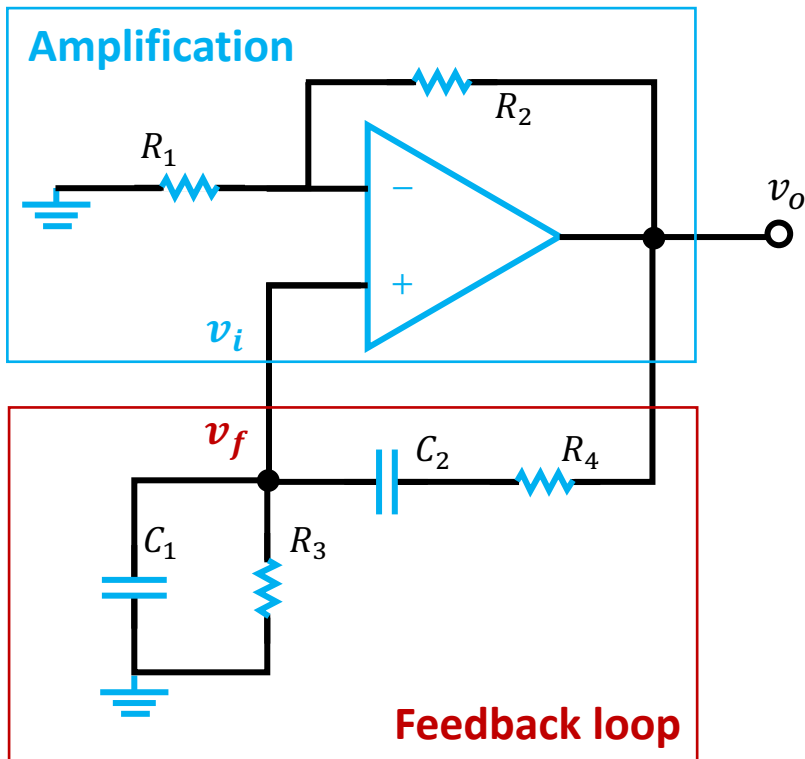
$$A = \frac{v_o}{v_i} = \frac{R_2 + R_1}{R_1}$$

- For the feedback loop

$$\begin{aligned}\beta = \frac{v_f}{v_o} &= \frac{R_3 \parallel \frac{1}{sC_1}}{R_4 + \frac{1}{sC_2} + R_3 \parallel \frac{1}{sC_1}} \\ &= \frac{1}{3 + sRC + \frac{1}{sRC}}\end{aligned}$$

# Example 4: the Wien-Bridge Oscillator

**QUESTION:** Find the functionality of the following circuit with  $R_3 = R_4 = R$ ,  $C_1 = C_2 = C$ .



- The transfer function

$$H(s) = A\beta = \frac{1 + \frac{R_2}{R_1}}{3 + sRC + \frac{1}{sRC}}$$

- $H(s)$  is a real number, when

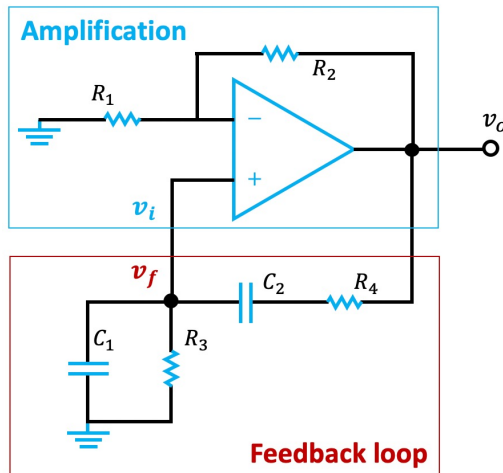
$$sRC + \frac{1}{sRC} = 0 \quad \Rightarrow \quad \omega_0 = \frac{1}{RC}$$

The oscillation frequency  $\omega_0$  is “determined” by the feedback loop

$$\begin{cases} A = \frac{v_o}{v_i} = \frac{R_2 + R_1}{R_1} \\ \beta = \frac{1}{3 + sRC + \frac{1}{sRC}} \end{cases}$$

# Example 4: the Wien-Bridge Oscillator

**QUESTION:** Find the functionality of the following circuit with  $R_3 = R_4 = R$ ,  $C_1 = C_2 = C$ .



- $A\beta = 1$  when  $\frac{R_2}{R_1} = 2$

**Positive feedback is observed**

- Oscillation is triggered when  $\frac{R_2}{R_1} > 2$

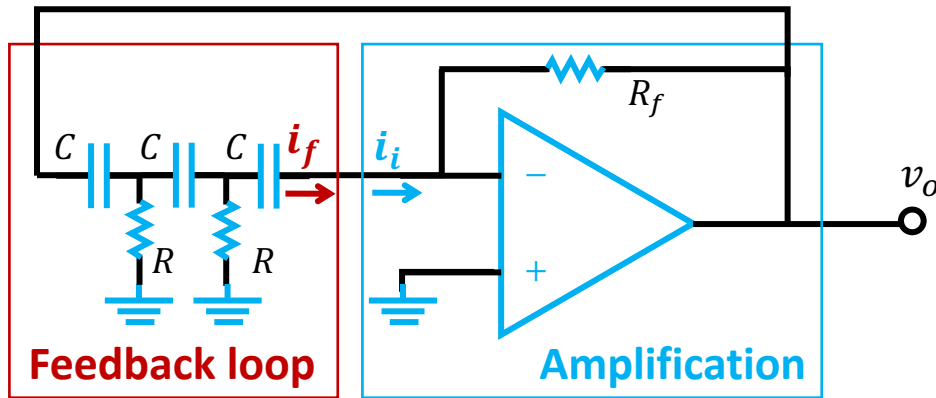
- The gain with feedback can be calculated as

$$\left\{ \begin{array}{l} A = \frac{v_o}{v_i} = \frac{R_2 + R_1}{R_1} \\ \beta = \frac{1}{3 + sRC + \frac{1}{sRC}} \\ H(s) = A\beta \text{ is real @ } \omega_0 = \frac{1}{RC} \end{array} \right.$$

$$A_f = \frac{A}{1 - A\beta} \xrightarrow{\omega = \omega_0} \frac{A}{1 - \frac{1}{3}A}$$

# Example 5: the Phase-Shift Oscillator

**QUESTION:** Find the functionality of the following



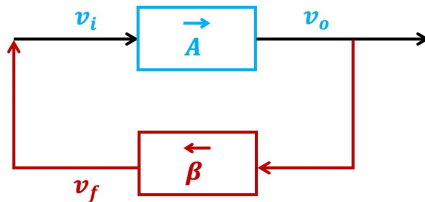
- For the amplification system, the gain

$$A = \frac{v_o}{i_i} = -R_f$$

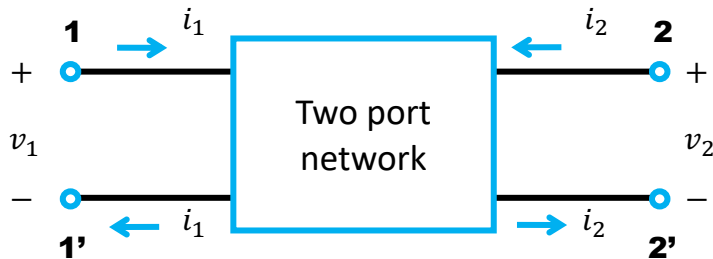
- For the feedback loop

$$\beta = \frac{i_f}{v_o}$$

Recall: op-amp w/  
positive feedback



# Recall: T parameters



- $v_1, i_1$  are dependent
- $v_2, i_2$  are independent

→ **T parameters**

$$\begin{cases} v_1 = Av_2 - Bi_2 \\ i_1 = Cv_2 - Di_2 \end{cases}$$

or

$$\begin{bmatrix} v_1 \\ i_1 \end{bmatrix} = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \begin{bmatrix} v_2 \\ -i_2 \end{bmatrix}$$

$$A = \frac{v_1}{v_2} \quad \text{when } i_2 = 0$$

$$B = -\frac{v_1}{i_2} \quad \text{when } v_2 = 0$$

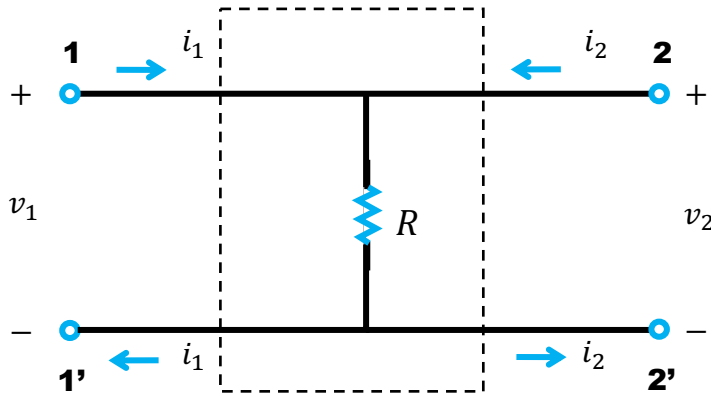
$$C = \frac{i_1}{v_2} \quad \text{when } i_2 = 0$$

$$D = -\frac{i_1}{i_2} \quad \text{when } v_2 = 0$$

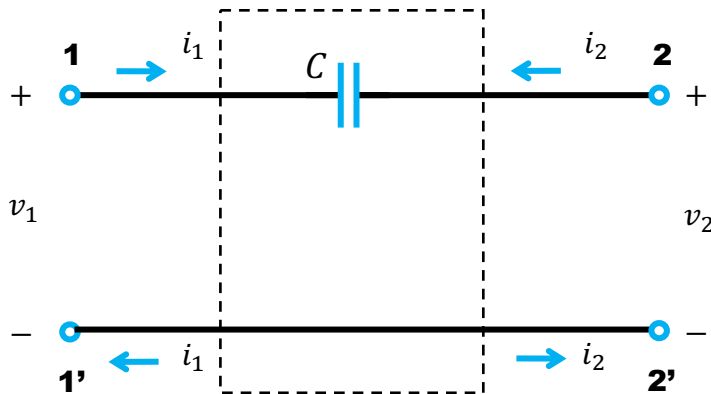
**T Parameters are also called as ABCD parameters**

# Recall: T parameters

**QUESTION:** find the  $T$  parameters of the circuit below



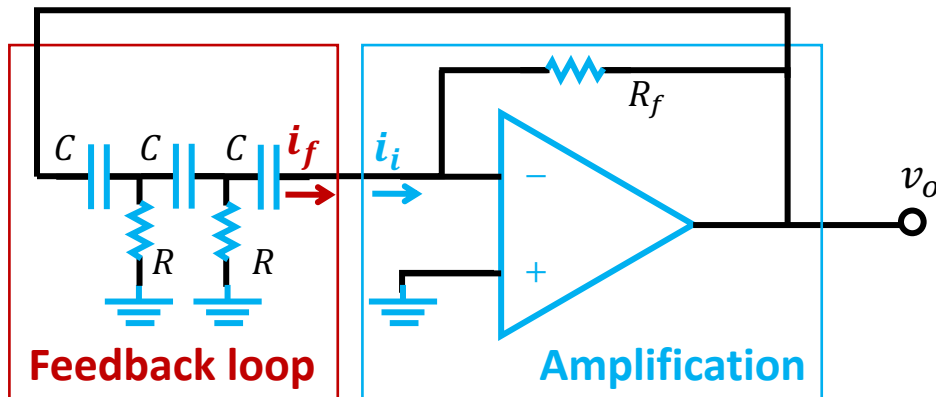
$$\begin{bmatrix} v_1 \\ i_1 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ \frac{1}{R} & 1 \end{bmatrix} \begin{bmatrix} v_2 \\ -i_2 \end{bmatrix}$$



$$\begin{bmatrix} v_1 \\ i_1 \end{bmatrix} = \begin{bmatrix} 1 & \frac{1}{sC} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v_2 \\ -i_2 \end{bmatrix}$$

# Example 5: the Phase-Shift Oscillator

**QUESTION:** Find the functionality of the following



- For the amplification system, the gain

$$A = \frac{v_o}{i_i} = -R_f$$

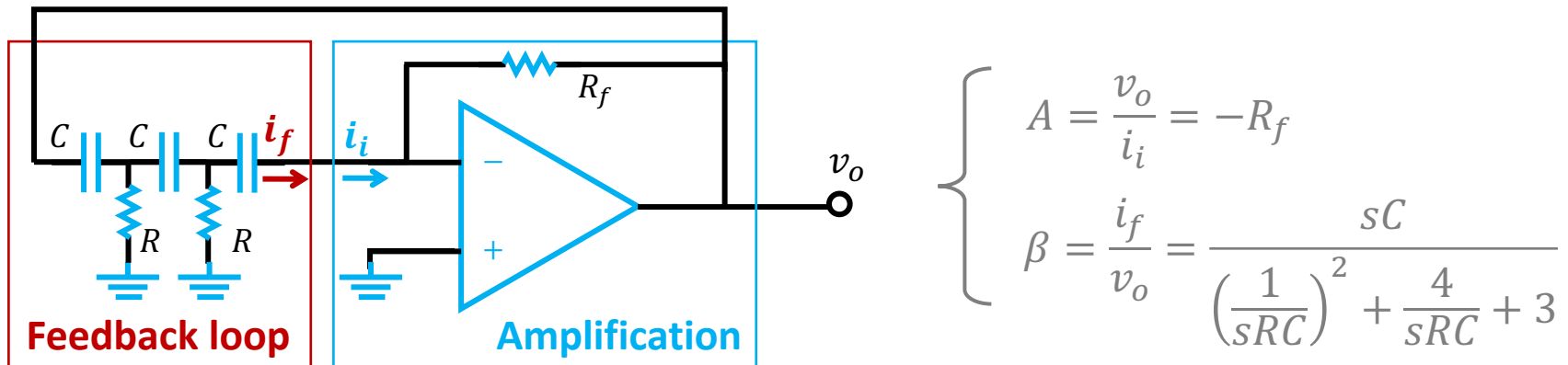
- For the feedback loop  $\beta = \frac{i_f}{v_o}$

$$\begin{bmatrix} v_o \\ i_o \end{bmatrix} = \begin{bmatrix} 1 & \frac{1}{sC} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ \frac{1}{R} & 1 \end{bmatrix} \begin{bmatrix} 1 & \frac{1}{sC} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ \frac{1}{R} & 1 \end{bmatrix} \begin{bmatrix} 1 & \frac{1}{sC} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v_f \\ i_f \end{bmatrix}$$

$$= \begin{bmatrix} * & \left[ \left( \frac{1}{sRC} \right)^2 + \frac{4}{sRC} + 3 \right] \frac{1}{sC} \\ * & * \end{bmatrix} \begin{bmatrix} v_f \\ i_f \end{bmatrix} \quad \Rightarrow \quad \beta = \frac{i_f}{v_o} = \frac{sC}{\left( \frac{1}{sRC} \right)^2 + \frac{4}{sRC} + 3}$$

# Example 5: the Phase-Shift Oscillator

**QUESTION:** Find the functionality of the following



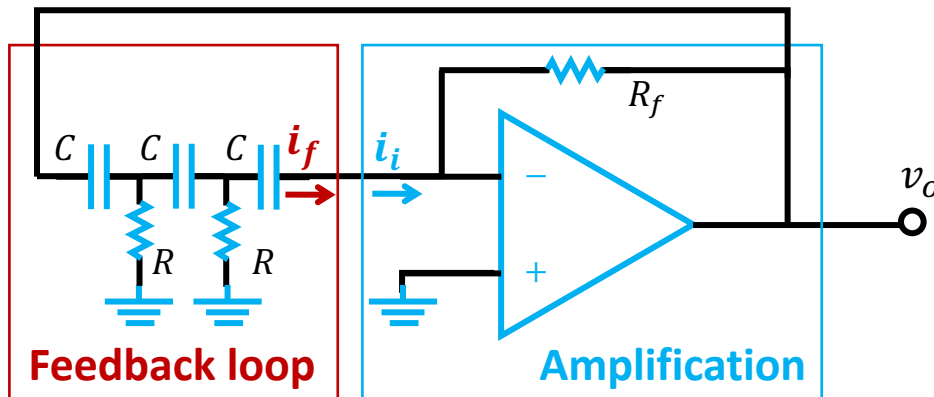
- The transfer function  $H(s) = A\beta = -\frac{sR_fC}{\left(\frac{1}{sRC}\right)^2 + \frac{4}{sRC} + 3} = |H(j\omega)|e^{j\varphi(\omega)}$

$$\text{where } \left\{ \begin{array}{l} |H(j\omega)| = \frac{R_f(\omega RC)^3}{R\sqrt{1 + 10(\omega RC)^2 + 9(\omega RC)^4}} \\ \varphi(\omega) = \frac{\pi}{2} - \arctan \frac{4\omega Rc}{1 - 3\omega^2 R^2 C^2} \end{array} \right.$$



# Example 5: the Phase-Shift Oscillator

**QUESTION:** Find the functionality of the following



$$\left\{ \begin{array}{l} A = \frac{v_o}{i_i} = -R_f \\ \beta = \frac{i_f}{v_o} = \frac{sC}{\left(\frac{1}{sRC}\right)^2 + \frac{4}{sRC} + 3} \\ H(s) = A|\beta(j\omega)|e^{j\varphi(\omega)} \end{array} \right.$$

- Positive feedback is observed when  $\left\{ \begin{array}{l} H(s) \text{ is a real number} \\ H(s) > 1 \end{array} \right. \quad \Rightarrow \quad \omega_0 = \frac{1}{\sqrt{3}RC}$

The oscillation frequency  $\omega_0$  is “determined” by the feedback loop

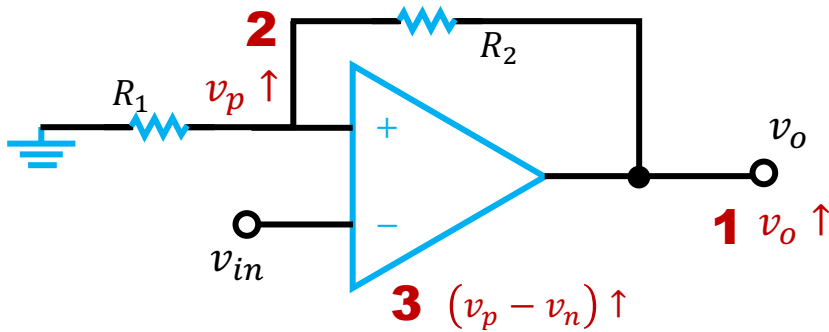
- Oscillation is triggered when  $A\beta(j\omega_0) > 1 \quad \Rightarrow \quad \frac{R_f}{12R} > 1$

# Outline

- HOW to generate an oscillation?
  - Negative resistance
  - Devices with negative resistance
  - Circuit features negative resistance
- Linear Oscillator
  - LC Oscillator
  - Crystal Oscillator
  - Op-Amp-RC Oscillator
- **Non-linear Oscillator**
  - **Bistable Circuit**

# Example 6: Schmitt Trigger

**QUESTION:** Find the functionality of the following circuit.



**THIS IS A POSITIVE FEEDBACK**

**The op-amp biased @ saturation region**

- If there is an increase @  $v_o$
- The noninverting input  $v_p$  increases correspondingly
- If the op-amp is biased in the linear region,  $v_o = A(v_p - v_n)$  increases

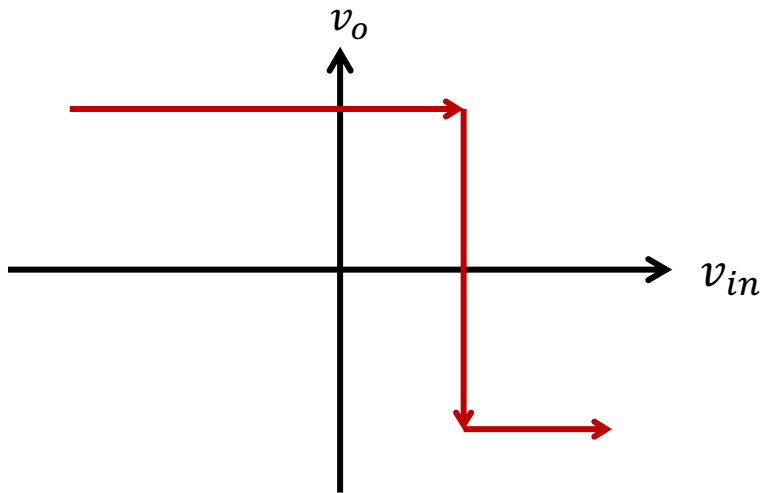
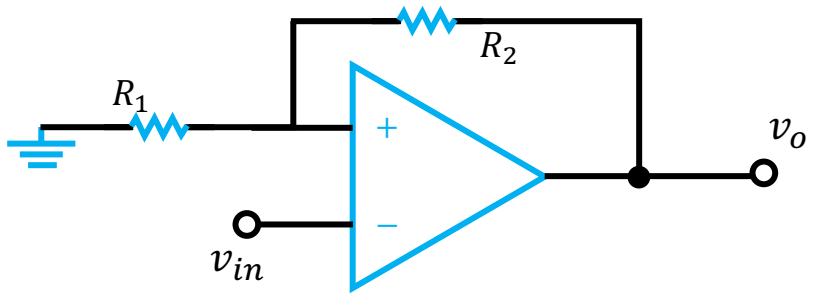
$v_o \uparrow$

$\updownarrow$

$v_o \uparrow$

# Example 6: Schmitt Trigger

**QUESTION:** Find the functionality of the following circuit.

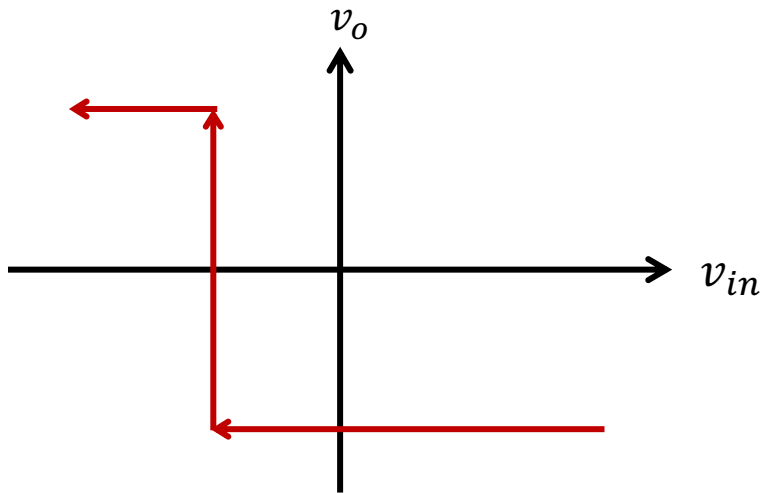
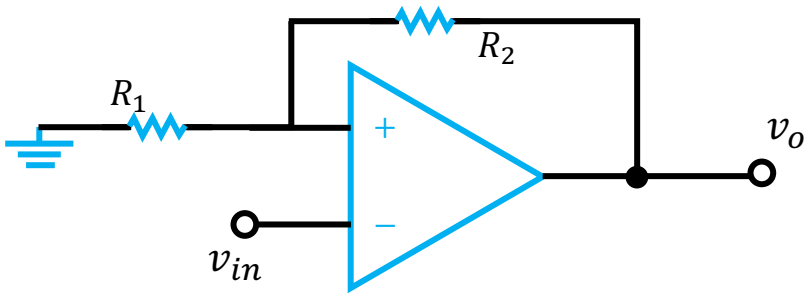


- If  $v_{in}$  is very small
- The op-amp is biased in positive saturation region,  $v_o = V_{sat}$
- According to KVL
$$v_+ = \frac{R_1}{R_1 + R_2} V_{sat}$$
- When  $v_{in} > v_+$ , the op-amp flips into the negative saturation region,  $v_o = -V_{sat}$
- According to KVL

$$v_+ = -\frac{R_1}{R_1 + R_2} V_{sat}$$

# Example 6: Schmitt Trigger

**QUESTION:** Find the functionality of the following circuit.



- If  $v_{in}$  is very large
- The op-amp is biased in negative saturation region,  $v_o = -V_{sat}$
- According to KVL

$$v_+ = -\frac{R_1}{R_1 + R_2} V_{sat}$$

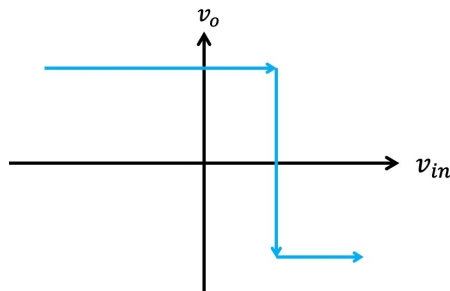
- When  $v_{in} < v_+$ , the op-amp flips into the positive saturation region,  $v_o = V_{sat}$
- According to KVL

$$v_+ = \frac{R_1}{R_1 + R_2} V_{sat}$$

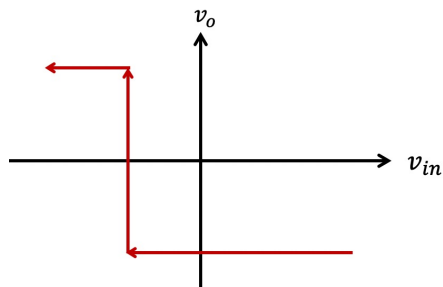
# Example 6: Schmitt Trigger

**QUESTION:** Find the functionality of the following circuit.

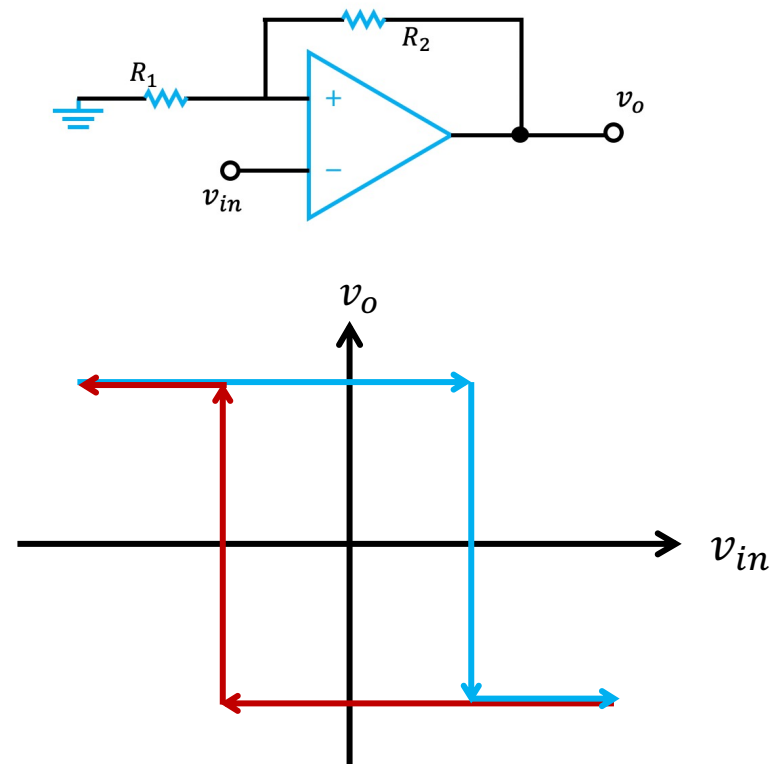
$v_{in}$  increases from a small value



$v_{in}$  decreases from a large value

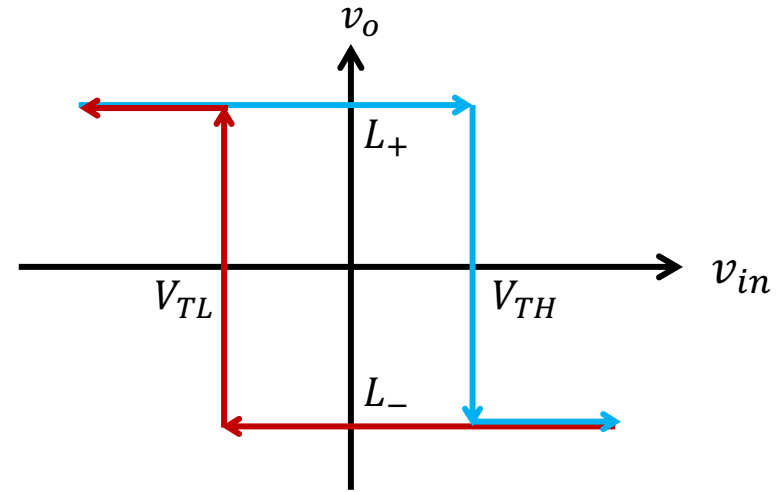
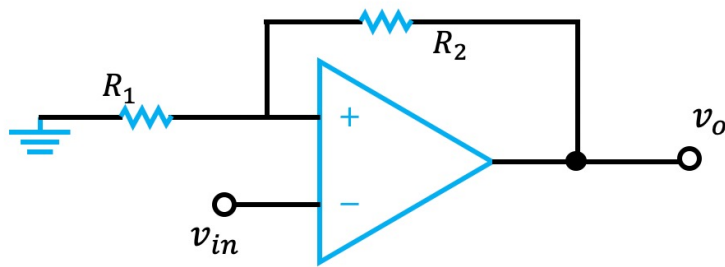


=



# Example 6: Schmitt Trigger

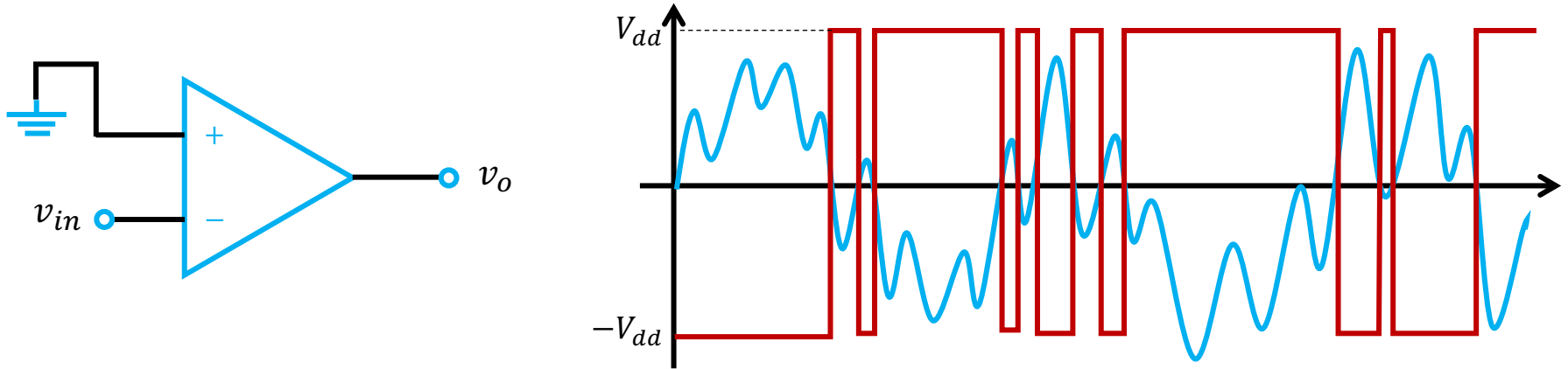
**QUESTION:** Find the functionality of the following circuit.



- Two possible levels:  $L_+$ ,  $L_-$
- Two thresholds:  $V_{TH}$ ,  $V_{TL}$

# Recall: zero-crossing detector

**QUESTION:** Find the output of the circuit. The op-amp is ideal.

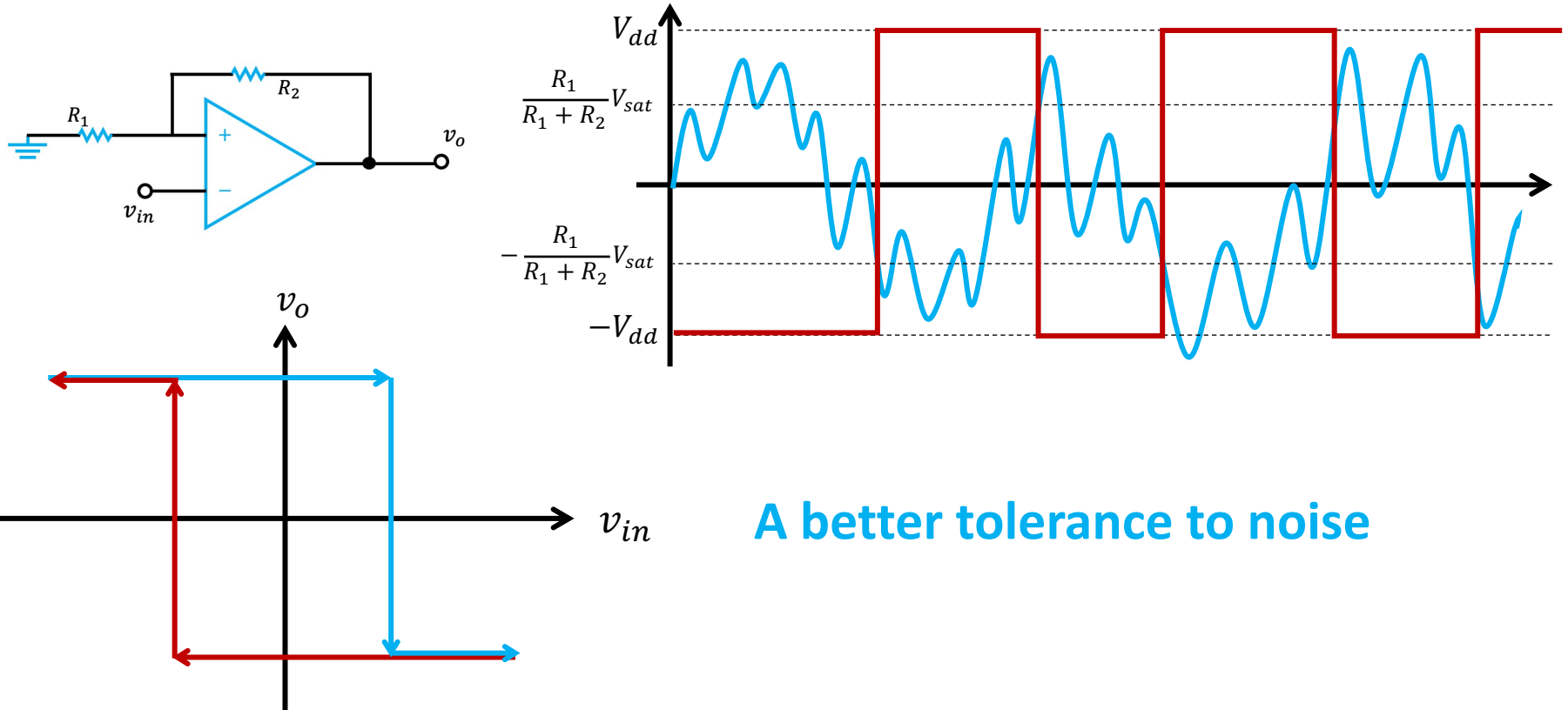


- If  $v_{in} > 0 \quad \rightarrow v_o = -V_{dd}$
- If  $v_{in} < 0 \quad \rightarrow v_o = V_{dd}$



# Example 6: Schmitt Trigger

**QUESTION:** Find the output of the circuit. The op-amp is ideal.

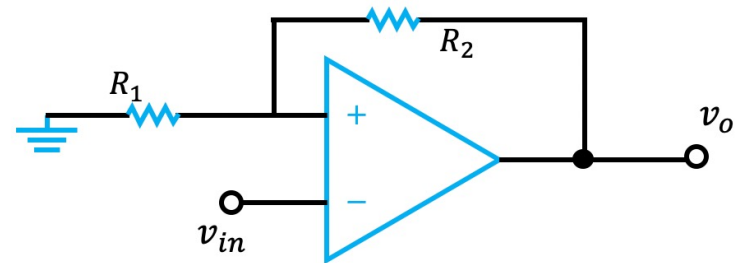


# Outline

- HOW to generate an oscillation?
  - Negative resistance
  - Devices with negative resistance
  - Circuit features negative resistance

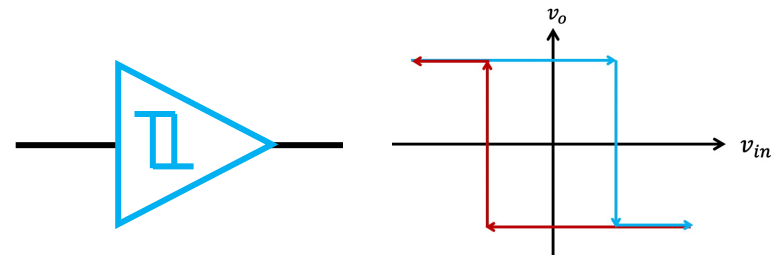
- Linear Oscillator

- LC Oscillator
  - Crystal Oscillator
  - Op-Amp-RC Oscillator



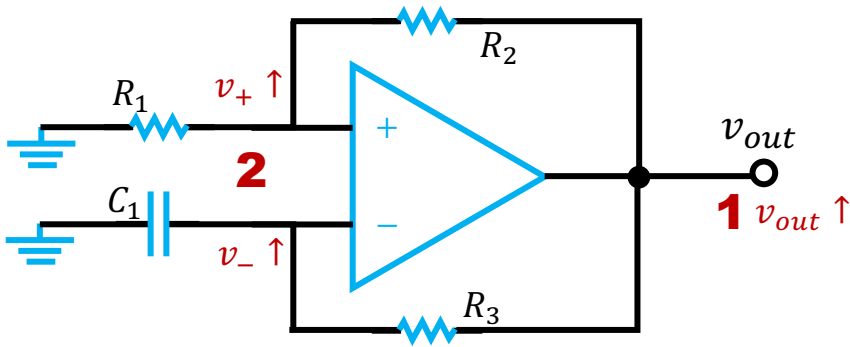
- Non-linear Oscillator

- Bistable Circuit
  - **Astable Multivibrators**



# Example 7

**QUESTION:** Find the functionality of the following circuit.



**What are  $R_2$  and  $R_3$  used for?**

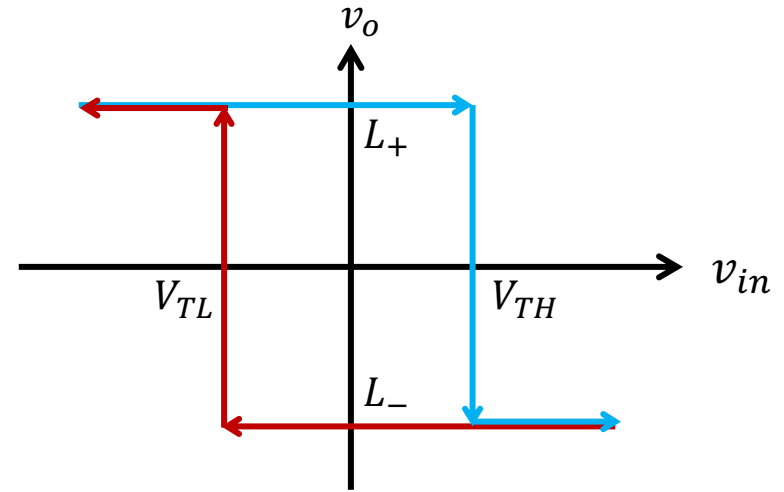
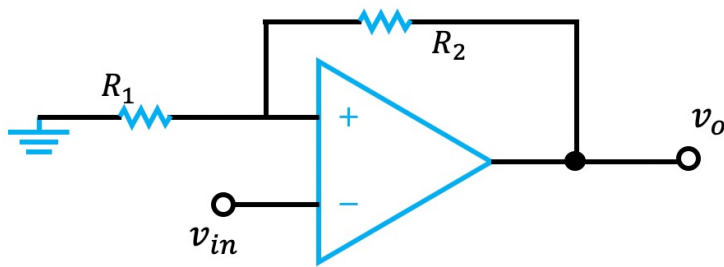
- If there is an increase @  $v_{out}$
- The input voltages  $v_+$  and  $v_-$  increase

○ If  $\Delta v_+ > \Delta v_-$   
 $v_{out} \uparrow$  causing an increasing of  $v_{out}$   
**POSITIVE FEEDBACK is observed**

○ If  $\Delta v_+ < \Delta v_-$   
 $v_{out} \downarrow$  causing an decreasing of  $v_{out}$   
**NEGATIVE FEEDBACK is observed**

# Recall: Schmitt Trigger

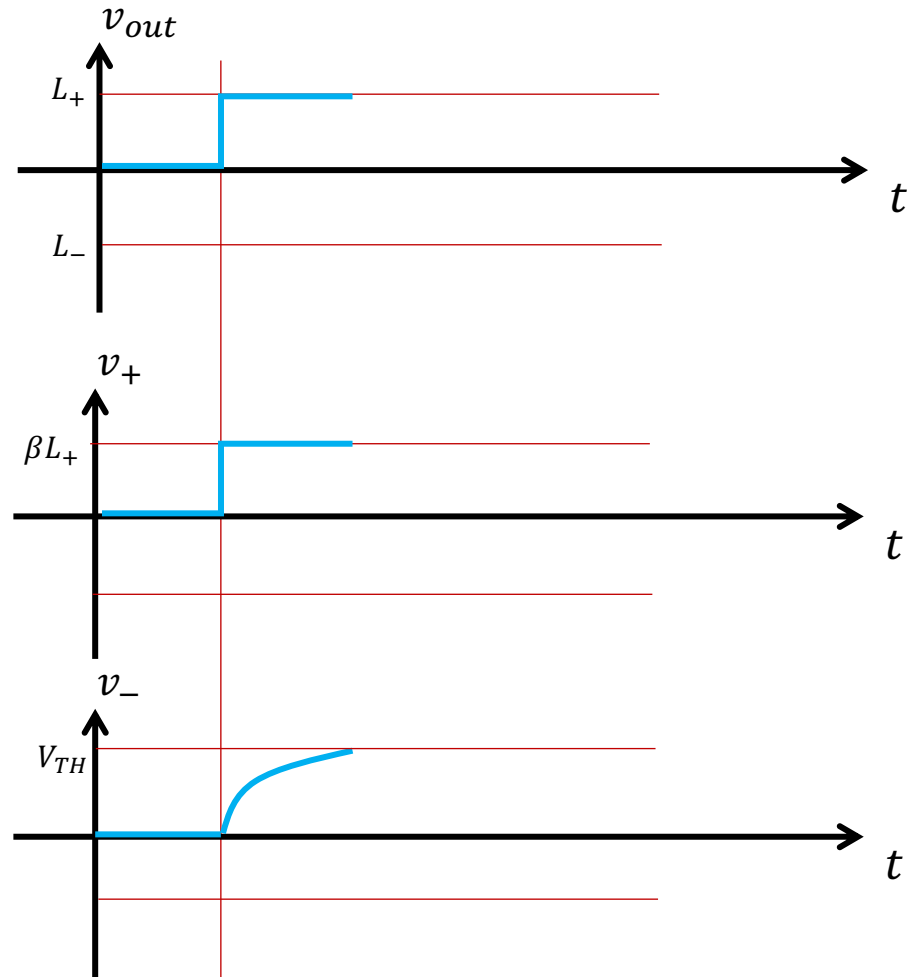
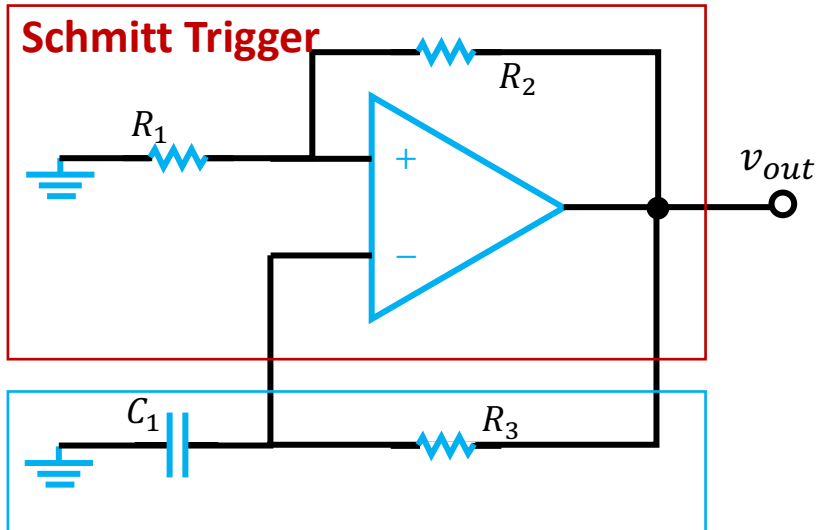
**QUESTION:** Find the functionality of the following circuit.



- Two possible levels:  $L_+$ ,  $L_-$
- Two thresholds:  $V_{TH}$ ,  $V_{TL}$

# Example 7

**QUESTION:** Find the functionality of the following circuit.



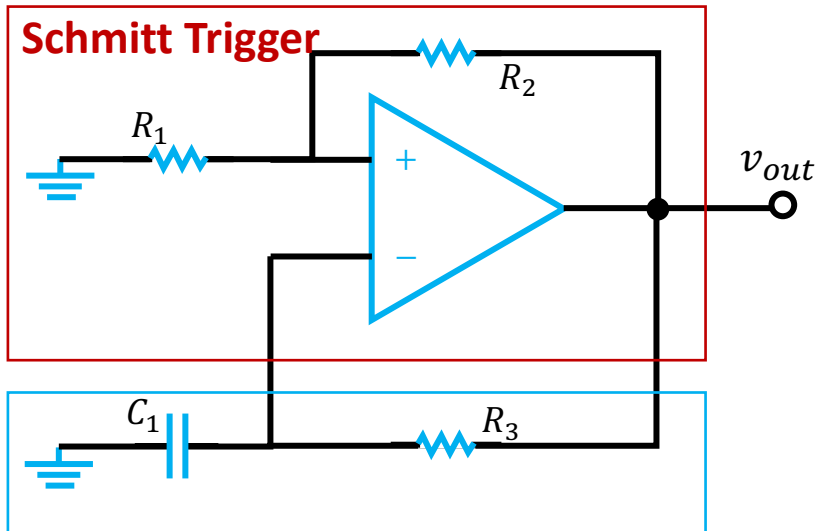
- Assume  $v_{out}$  flipped to  $L_+$  due to unexpected disturb
- According to KVL

$$v_+ = \frac{R_1}{R_1 + R_2} L_+ = \beta L_+$$

- $C_1$  is charged to  $V_{TH}$

# Example 7

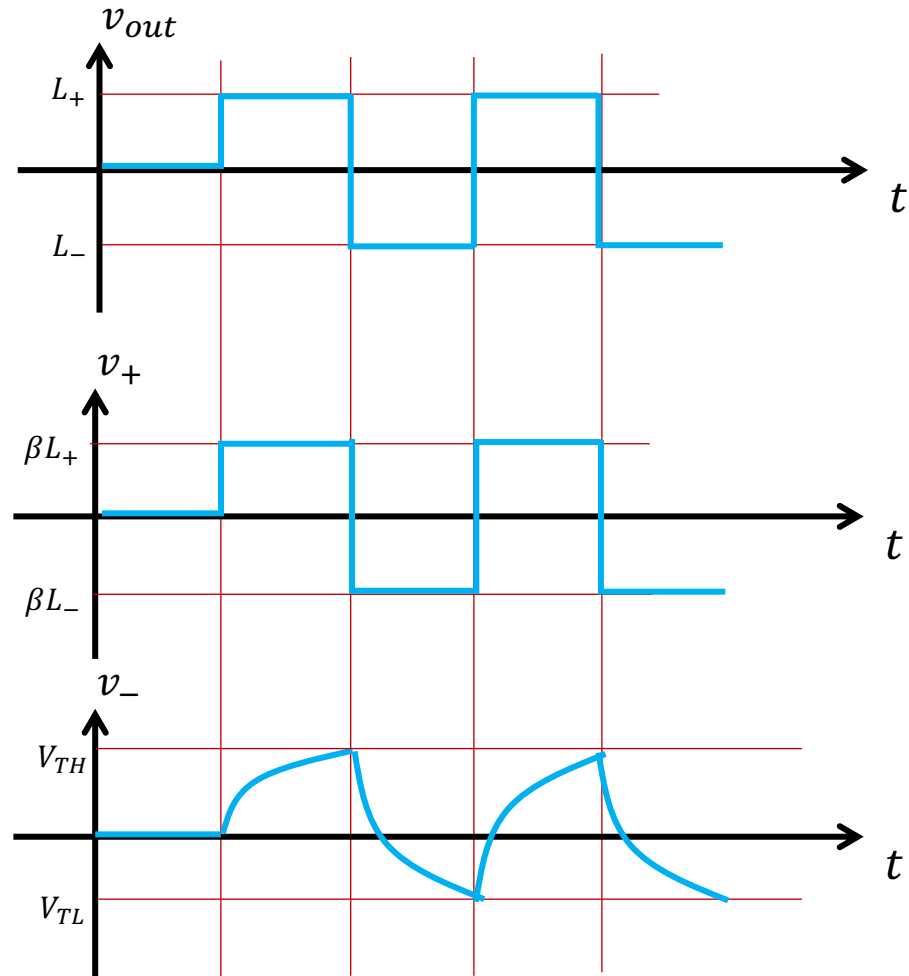
**QUESTION:** Find the functionality of the following circuit.



- $v_{out}$  flipped to  $L_-$  due to the increasing of the  $v_-$
- According to KVL

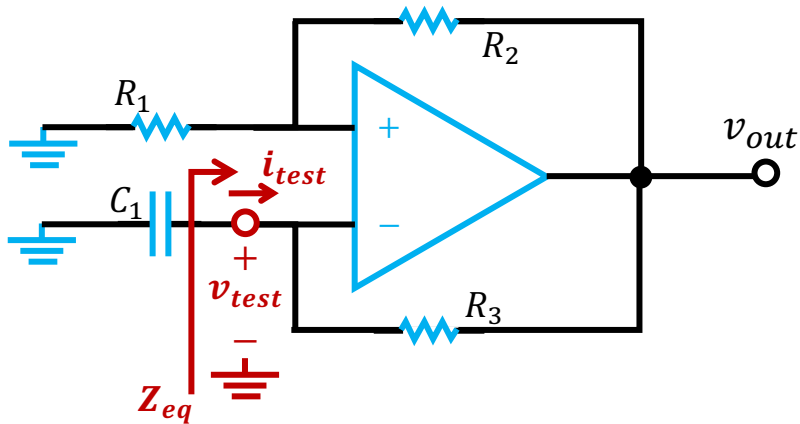
$$v_+ = \frac{R_1}{R_1 + R_2} L_- = \beta L_-$$

- $C_1$  is discharged to  $V_{TL}$



# Example 7

**QUESTION:** Find the functionality of the following circuit.



- If the op-amp is biased in **linear region**

Approximately  $v_+ = v_-$

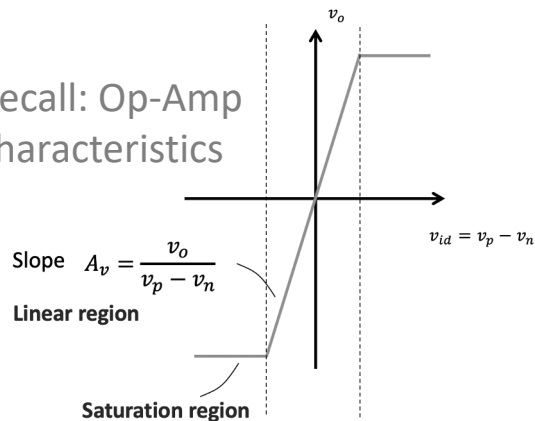
- According to KVL

$$v_+ = v_- = \frac{R_1}{R_1 + R_2} v_{out}$$

$$i_{test} = \frac{v_{test} - v_{out}}{R_3}$$

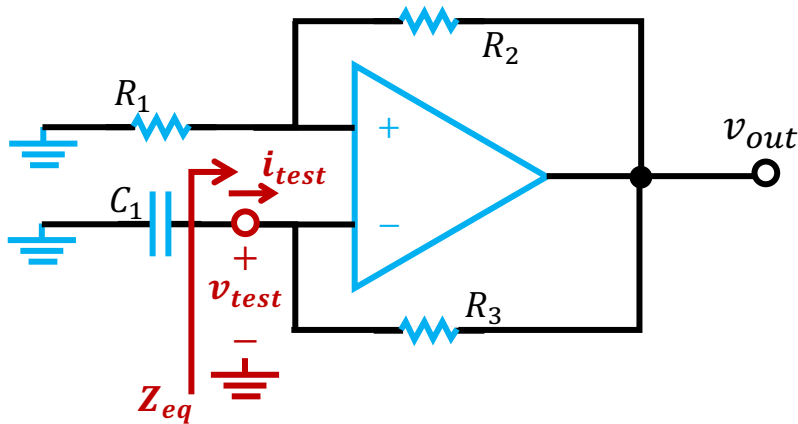
$$Z_{eq} = \frac{v_{test}}{i_{test}} = \frac{v_-}{i_{test}} = -\frac{R_1}{R_2} R_3$$

Recall: Op-Amp characteristics



# Example 7

**QUESTION:** Find the functionality of the following circuit.



- If the op-amp is biased in **saturation region**
- If  $v_{out} = -V_{sat}$  ( $v_+ \ll v_-$ )
- According to KVL

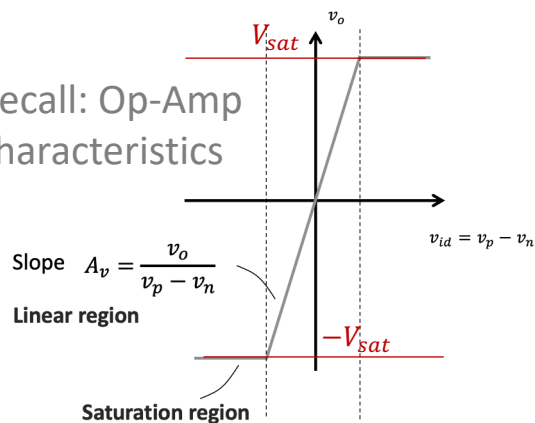
$$i_{test} = \frac{v_{test} - v_{out}}{R_3} = \frac{v_{test} + V_{sat}}{R_3}$$

$$Z_{eq} = \frac{v_{test}}{i_{test}} = R_3 \frac{1}{1 + \frac{V_{sat}}{v_{test}}}$$

- To keep  $v_{out} = -V_{sat}$

$$v_- > v_+ = -\frac{R_1}{R_1 + R_2} V_{sat}$$

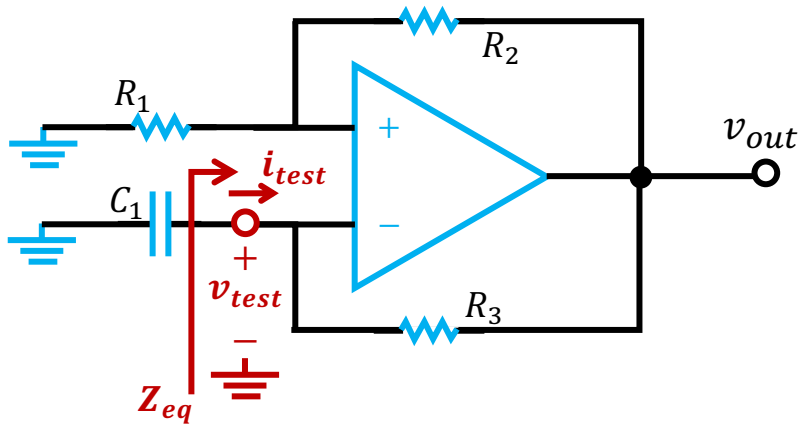
Recall: Op-Amp characteristics





# Example 7

**QUESTION:** Find the functionality of the following circuit.



- If the op-amp is biased in **saturation region**
- If  $v_{out} = V_{sat}$
- According to KVL

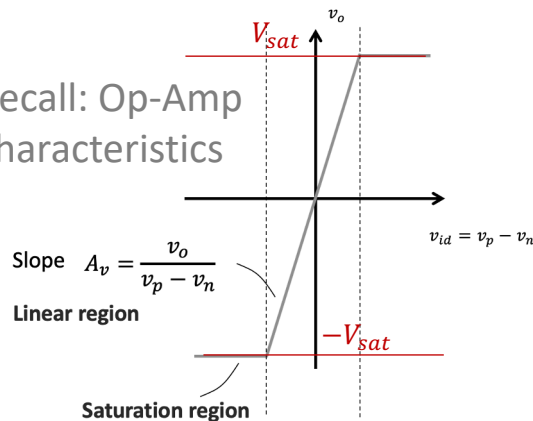
$$i_{test} = \frac{v_{test} - v_{out}}{R_3} = \frac{v_{test} - V_{sat}}{R_3} < 0$$

$$Z_{eq} = \frac{v_{test}}{i_{test}} = R_3 \frac{1}{1 - \frac{V_{sat}}{v_{test}}}$$

- To keep  $v_{out} = V_{sat}$

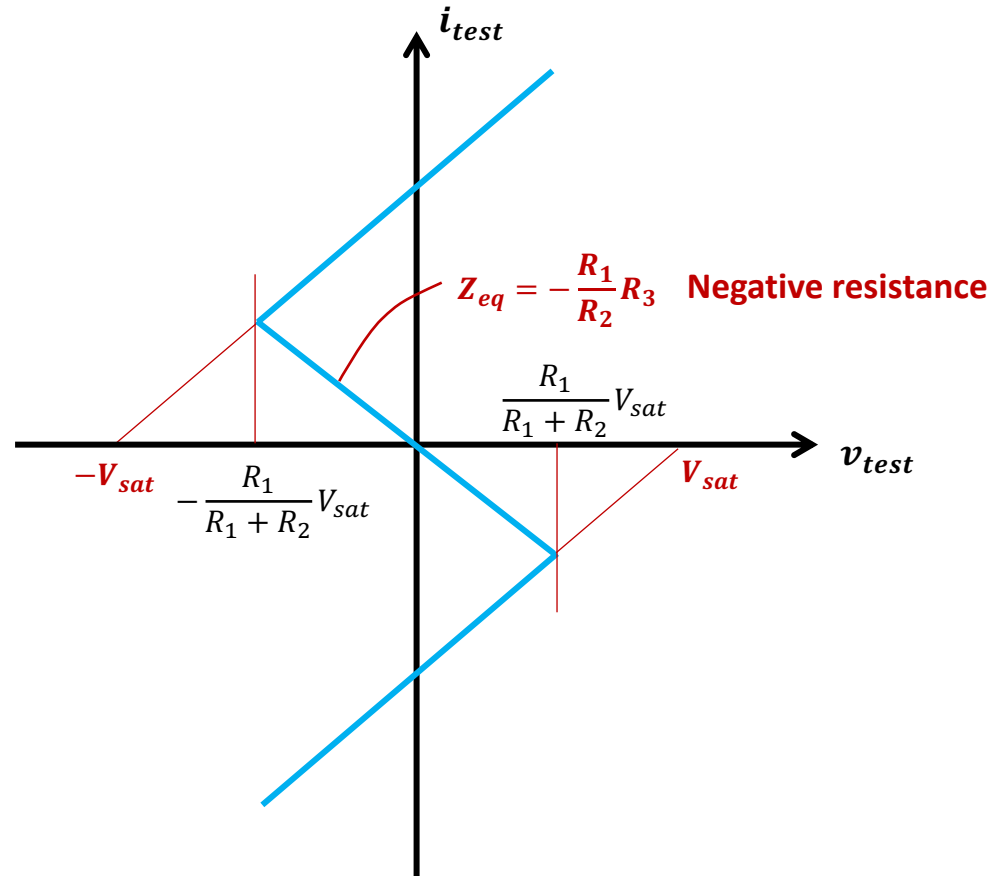
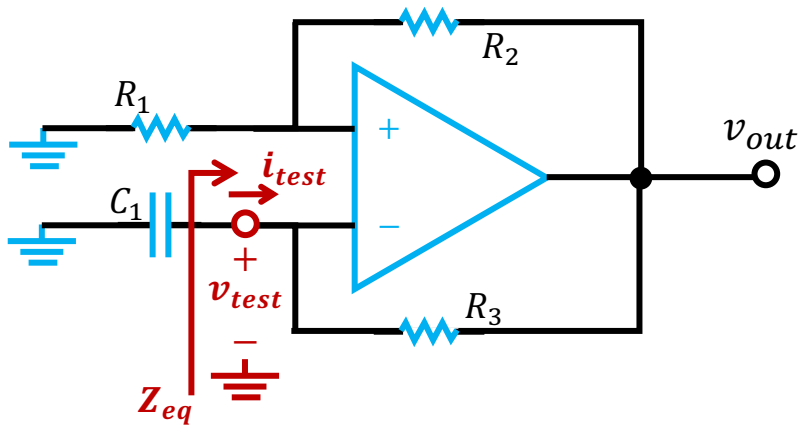
$$v_- < v_+ = \frac{R_1}{R_1 + R_2} V_{sat}$$

Recall: Op-Amp characteristics



# Example 7

**QUESTION:** Find the functionality of the following circuit.



Recall: Op-Amp characteristics

$$\text{Slope } A_v = \frac{v_o}{v_p - v_n}$$

Linear region

Saturation region

# Outline

- HOW to generate an oscillation?
  - Negative resistance
  - Devices with negative resistance
  - Circuit features negative resistance
- Linear Oscillator
  - LC Oscillator
  - Crystal Oscillator
  - Op-Amp-RC Oscillator
- Non-linear Oscillator
  - Bistable Circuit
  - Astable Multivibrators

# Reading tasks & learning goals

- Reading tasks

- Microelectronic Circuits, 6<sup>th</sup> edition
  - Chapter 17.1 – 17.6

- Learning goals

- Well understand the concept of **negative resistance**
- Well understand how to calculate the specification of a **linear oscillation circuit**
- Well understand how to analyze a circuit consisting of **op-amp with positive feedback**
- Well understand how to calculate the specification of a **non-linear oscillation circuit**